Critical Analysis of Quantum-Inspired Evolutionary Algorithms for Multi-Objective Optimization: Taxonomy, Features, and Limitations

Olivian-Antonio Gîlea Adrian Florea

October 10, 2026

Abstract

Quantum-inspired evolutionary algorithms (QIEAs) borrow the Q-bit chromosome and rotation-gate operator from quantum mechanics to improve classical evolutionary search. Their multi-objective optimization (MOO) branch, active since at least 2003, remains sparse, scattered across domains, and, to our knowledge, unsurveyed on its own terms. A bibliometric analysis of 2,250 QIEA papers from 1996–2026 shows multi-objective work making up under a tenth of the field, with its share roughly doubling since 2024, and clustering with application domains rather than with algorithm design. We propose a taxonomy of QIEA-for-MOO methods along five axes (representation, operator type, selection mechanism, hybridization, and application domain), apply it in a feature comparison that uses single-objective QIEAs as a baseline, and critically analyze the field's limitations. Across thirteen methods selected by citation, selection mechanisms are varied but never reference-point- or hypervolume-based, several highly cited methods do not state how they combine objectives, and methods are rarely benchmarked against one another. The rotation gate's reference solution, obvious with one objective, changes most with several, yet only three methods document their choice. Reproducibility gaps, weak benchmarking, and scalability claims tested at a single problem size recur across method categories. An illustrative case study, a block-partitioned quantum-circuit-optimization pipeline extending a published NSGA-II framework with NSGA-III and SMS-EMOA, shows how an adaptive, per-run hypervolume reference point can change a cross-algorithm ranking and how a correctness bug can silently invalidate weeks of results. We close with open challenges, including the untested combination of a multi-objective QIEA with a quantum-computing problem domain.

1 Introduction

Quantum computing has progressed from theoretical models of computation to experimentally realized, programmable hardware, yet the currently available noisy intermediate-scale quantum (NISQ) devices remain limited in qubit count, connectivity, and coherence time [1]. This gap between the algorithms quantum computing ultimately promises and the hardware available today has created a specific, practical need: classical heuristics capable of tuning, compiling, and optimizing quantum circuits and quantum-algorithm parameters well enough that near-term devices can run them at all. Evolutionary algorithms (EAs) are a natural fit for this role – population-based, derivative-free, and already used to design and compress circuits directly, as this survey's own case study (Section 7) illustrates.

A second, independent thread of research asks the opposite question: can ideas *from* quantum mechanics – superposition, interference, entanglement – improve classical evolutionary algorithms themselves, on classical hardware, whether or not the resulting solutions ever touch a real qubit? This is the domain of *quantum-inspired evolutionary algorithms* (QIEAs), founded on Han and Kim's Q-bit chromosome and Q-gate rotation operator [2] and since applied across combinatorial and continuous optimization alike [1, 3]. Following Rehman et al. [1], we adopt the terminology that separates *quantum-inspired* evolutionary algorithms (QiEA: classical algorithms inspired by quantum phenomena) from *quantum* evolutionary algorithms (QEA: algorithms designed for and executed on physical quantum hardware). This survey is scoped to the former.

Whether classical or quantum-inspired, evolutionary algorithms are above all optimization methods – yet the optimization problems they are applied to seldom reduce to a single objective. Most real engineering and scientific problems are not single-objective: they require trading off several competing criteria (e.g. fidelity against circuit depth, or cost against coverage) rather than optimizing one metric in isolation. Multi-objective optimization (MOO) has its own mature toolbox of classical evolutionary methods – NSGA-II [4], NSGA-III [5], SPEA2 [6], MOEA/D [7], and SMS-EMOA [8] chief among them – and QIEA research has followed MOO into this territory since at least Zhang et al. [9]'s quantum evolutionary algorithm for multi-objective digital-filter design, with Pareto-based methods such as Kim et al. [10]'s multi-objective knapsack algorithm following shortly after. Yet the literature applying QIEA specifically to multi-objective problems remains comparatively sparse and scattered across domains [11–13], compared to the much larger body of QIEA work on single- or bi-objective problems such as feature subset selection [3, 14]. Part of the reason lies in the representation itself: the Q-bit chromosome encodes each individual as a vector of qubits that is observed into a binary string, a natural fit for binary and combinatorial problems [2] but not for problems whose solutions are more

naturally encoded as permutations or real-valued vectors, where an additional decoding step is required. As a result, QIEAs do not always match classical EAs that operate on such encodings directly: Olvera et al. [11], adapting two originally combinatorial QIEA-MOO algorithms to continuous search spaces, found that floating-point NSGA-II and SMPSO came closer to the true Pareto front owing to their precision advantage, and that NSGA-II converged faster overall, even though the quantum-inspired variants were statistically competitive – and that the quantum-inspired variants' performance degraded as the number of decision variables, and with it the number of Q-bits, increased. No survey, to our knowledge, systematically taxonomizes and critically compares QIEA-for-MOO methods on their own terms. This is the gap this survey addresses.

Beyond the mechanics of any one algorithm, the appeal of cheaper, shallower, more efficient optimization has a stake that outlives any individual benchmark result: computing's energy footprint. Large-scale classical compute – from generative-AI training runs to combinatorial optimization on classical hardware – carries real energy and sustainability costs [15], and reducing the resources a given optimization problem requires (fewer generations, cheaper fitness evaluations, shallower resulting circuits) is not merely a performance metric but part of a broader case for energy-efficient, sustainable computing [16]. Optimization-driven resource management already delivers this kind of saving concretely outside quantum computing – reinforcement-learning-based energy management in community microgrids, for instance, has been shown to meaningfully cut both cost and carbon emissions relative to rule-based control [17] – and the same underlying argument, that a cheaper optimization process is itself a sustainability gain, applies to this survey's subject matter. This is one of the underlying reasons QIEA-for-MOO is worth taking seriously as its own subfield rather than as a footnote to single-objective QIEA or to classical MOO.

To ground the survey's more abstract taxonomy and critical-analysis claims in a concrete example, Section 7 presents a case study built on this project's own quantum-circuit-optimization pipeline, which partitions a target circuit into qubit blocks, evolves a cheaper per-block circuit via a pluggable multi-objective optimizer, and reassembles the result. That pipeline is itself a direct extension of Ghlib et al. [18]'s scalable, block-partitioned NSGA-II framework for quantum circuit optimization, adding NSGA-III and SMS-EMOA as alternative per-block optimizers. The case study is presented as an illustrative worked example, not as this survey's primary evidence; full empirical validation of that pipeline is the subject of a separate, forthcoming empirical results paper.

1.1 Scope and Inclusion Criteria

This survey covers literature at the intersection of quantum-inspired evolutionary computation and multi-objective optimization, identified through a combination of targeted literature search (arXiv, IEEE Xplore, Springer, ACM Digital Library, MDPI) and citation-chasing from the papers found, spanning publications from 1985 (the earliest Pareto-based MOEA, VEGA [19]) to 2026. This targeted selection is complemented by a bibliometric sweep of the QIEA literature as a whole, 2,250 papers from 1996–2026 retrieved from OpenAlex, which also contributes its ten most-cited papers to the survey (Section 2). We include:

- Papers proposing or surveying quantum-inspired evolutionary algorithms (in the QiEA sense above) applied to multi-objective or many-objective optimization problems, on continuous or combinatorial search spaces.
- Classical (non-quantum-inspired) multi-objective evolutionary algorithms that QIEA-MOO methods are built on, hybridized with, or benchmarked against – including the full Pareto-dominance lineage (VEGA through SPEA2 and beyond), indicator-based methods, and decomposition-based methods – since these form the baseline vocabulary the taxonomy in Section 4 depends on.
- Single-objective QIEAs, as a comparison baseline: the foundational representation and operator papers [2, 20, 21] and broad single-objective QIEA surveys [3, 14]. These are included not as QIEA-MOO methods in their own right but to show what changes, and what stays the same, when a QIEA moves from one objective to several (Sections 4.1 and 6.2).
- Papers on (multi-objective) genetic algorithms for quantum circuit optimization specifically, since these motivate and inform the case study in Section 7.
- A small number of papers establishing the broader motivation for energy-efficient and sustainable computing, cited only in this introduction and the conclusion, and not treated as QIEA/MOO-technical literature.

We exclude quantum evolutionary algorithms designed to run on physical quantum hardware (QEA in the strict sense), except where a paper explicitly discusses the QiEA/QEA distinction. Only English-language sources are included.

The multi-objective methods that the taxonomy (Section 4) and feature comparison (Section 5) examine in depth are chosen by a stated rule rather

than by hand. They are the ten most-cited papers in the bibliometric corpus of Section 2 that are tagged multi-objective and whose own optimization method is a QIEA – skipping papers that use a QIEA only as a comparison baseline, and papers without an English full text – [10, 22–30], plus three recent methods found by the targeted search that a citation-based rule would miss because they have had little time to be cited [11–13]. This gives thirteen methods. For most of them, the comparison rests on what the paper's abstract states; Section 5 marks what could not be confirmed.

1.2 Research Questions and Contributions

This survey is organized around four research questions:

RQ1: How can quantum-inspired evolutionary algorithms for multi-objective optimization be systematically classified?

RQ2: How do the surveyed methods compare in representation, operators, selection mechanism, complexity, scalability, and reported performance?

RQ3: What are the main limitations of the QIEA-for-MOO literature, both per method category and across the field as a whole?

RQ4: How do these limitations manifest in a concrete multi-objective evolutionary pipeline applied to a quantum problem domain?

In answering them, this survey makes the following contributions, one per research question, preceded by a bibliometric overview of the field:

- A bibliometric analysis of 2,250 QIEA papers from 1996–2026: the field's publication trend, the share and recent growth of multi-objective work, its keyword clusters, and its ten most-cited papers (Section 2).
- **(RQ1)** A taxonomy of QIEA-for-MOO methods along five axes – representation scheme, quantum-inspired operator type, selection/diversity mechanism, hybridization with classical MOEAs, and application domain (Section 4).
- **(RQ2)** A systematic feature comparison of the surveyed methods against that taxonomy, covering representation, operator type, selection mechanism, and application domain alongside complexity, scalability, convergence properties, and reported performance (Section 5).
- **(RQ3)** A critical analysis of per-category weaknesses and cross-cutting open problems in the field – reproducibility gaps, weak benchmarking practices, and scalability claims that are rarely empirically stress-tested (Section 6).

- **(RQ4)** A worked case study that grounds several of these abstract claims in a concrete block-partitioned quantum-circuit-optimization pipeline, itself an extension of prior published work [18] (Section 7).

1.3 Paper Organization

Section 2 gives a bibliometric overview of the QIEA literature. Section 3 reviews the quantum-computing and multi-objective-optimization background needed to follow the rest of the survey, including the classical MOEA baselines QIEA methods are compared against. Section 4 introduces the classification axes used throughout. Section 5 applies that taxonomy systematically across the surveyed literature. Section 6 discusses limitations and open problems, per taxonomy category and across the field as a whole. Section 7 presents the illustrative case study. Section 8 synthesizes open challenges and future directions, and Section 9 concludes.

2 Bibliometric Analysis

Before turning to individual methods, this section maps the QIEA literature as a whole: how large it is, how it has grown over the last 30 years, how much of it is multi-objective, which themes it clusters into, and which papers have been most influential. It complements the targeted selection of Section 1.1: that selection decides which methods the taxonomy examines in depth, while this section gives the field-level context in which to read them.

2.1 Data and Method

We queried OpenAlex [31], a fully open index of scholarly works covering all major publishers, on 3 October 2026. Two queries were run against titles and abstracts, restricted to publication years 1996–2026:

- **QIEA query:** `"quantum-inspired evolutionary" OR "quantum-inspired genetic" OR "quantum evolutionary algorithm" OR "quantum genetic algorithm" OR QIEA`. This is the corpus analysed in the rest of this section.
- **Broad query:** `("evolutionary computation" OR "evolutionary algorithm" OR "genetic algorithm") AND ("quantum computing" OR "quantum computer")`. This captures the wider intersection of evolutionary computation and quantum computing, including the second research thread of Section 1 (evolutionary algorithms applied *to* quantum problems), and is used only for the publication trend.

Each result set was then cleaned in four steps. (1) Only peer-reviewed document types were kept (journal articles, conference papers, book chapters, books, reviews, and dissertations), dropping preprints, software, and

datasets; this removed, among others, several dozen near-duplicate repository uploads dated 2026. (2) Only English-language records were kept, matching Section 1.1. (3) Each record's title and abstract, as returned by OpenAlex, was checked locally for the query terms. OpenAlex also matches records on text it does not return, such as some publisher-withheld abstracts, and this step removed off-topic matches (e.g. quantum-dot physics papers) at the cost of also dropping some genuine records whose abstract is not available. (4) Records with the same normalized title were merged, keeping the most-cited version. The QIEA query went from 2,835 records to 2,654, 2,603, 2,341, and finally 2,250 papers; the broad query from 935 to 558. Two publication years were corrected by hand where OpenAlex disagreed with the paper's own DOI (Narayanan and Moore [20], dated 2002 in OpenAlex, and Han et al. [32], also dated 2002).

A paper was tagged multi-objective if its title, abstract, or keywords mention multi-, many-, bi-, two-, three-, or four-objective optimization or Pareto optimality. Keyword clusters were built from the keywords OpenAlex assigns to each paper: synonyms were merged, the query terms themselves (present in nearly every record) were removed, and keywords occurring in at least 20 papers were kept. This gave 55 keywords, linked whenever two of them occur in the same paper and weighted by the number of such papers (695 links). Clusters were found with the Louvain method [33]. Citation counts are as reported by OpenAlex on the retrieval date. The full pipeline (queries, cleaning, figures, and tables) is a single script, `bibliometric_analysis.py`, distributed with the paper together with the raw query results it was run on, so every number in this section can be regenerated.

2.2 Publication Trend

Figure 1(a) shows the QIEA corpus by year. After Narayanan and Moore [20]'s 1996 paper, the field is almost empty until Han and Kim's papers of 2001–2004 [2, 21, 32], then grows quickly, from 24 papers in 2004 to a peak of 166 in 2010 and a second peak of 164 in 2013. Output then falls to 74 papers in 2017 before recovering to 135 in 2025; with 130 papers by early October, 2026 is on course to exceed it.

Multi-objective work is a small but growing part of this corpus: 215 of the 2,250 papers (9.6%) are tagged multi-objective. The share was 7.6% up to 2013 and 8.7% over 2014–2023, then roughly doubled to 16.6% over 2024–2026 (61 of 367 papers). The recent recovery of the field as a whole is therefore disproportionately multi-objective, which is part of the case for a survey focused on QIEA-for-MOO specifically.

The broad query (Figure 1(b)) follows a different curve: between 7 and 24 papers a year over 2005–2020, then a sharp rise from 2021, reaching 63 papers in 2024 and 99 in 2025. The 2026 figure (40 papers) is partial and should not be read as a decline. The composition of this corpus changes

with its growth. Over 2005–2020, 138 of its 241 papers (57%) also belong to the QIEA corpus, and only 24 (10%) mention quantum circuits, variational algorithms, QAOA, or similar quantum problems in their title or abstract. Over 2021–2026, only 97 of 294 (33%) are QIEA papers, while 87 (30%) mention such quantum problems. The recent growth at the intersection of evolutionary computation and quantum computing therefore comes mostly from evolutionary algorithms applied *to* quantum problems, the thread this survey's case study belongs to, rather than from quantum-inspired algorithms, and coincides with the availability of the NISQ hardware discussed in Section 1.

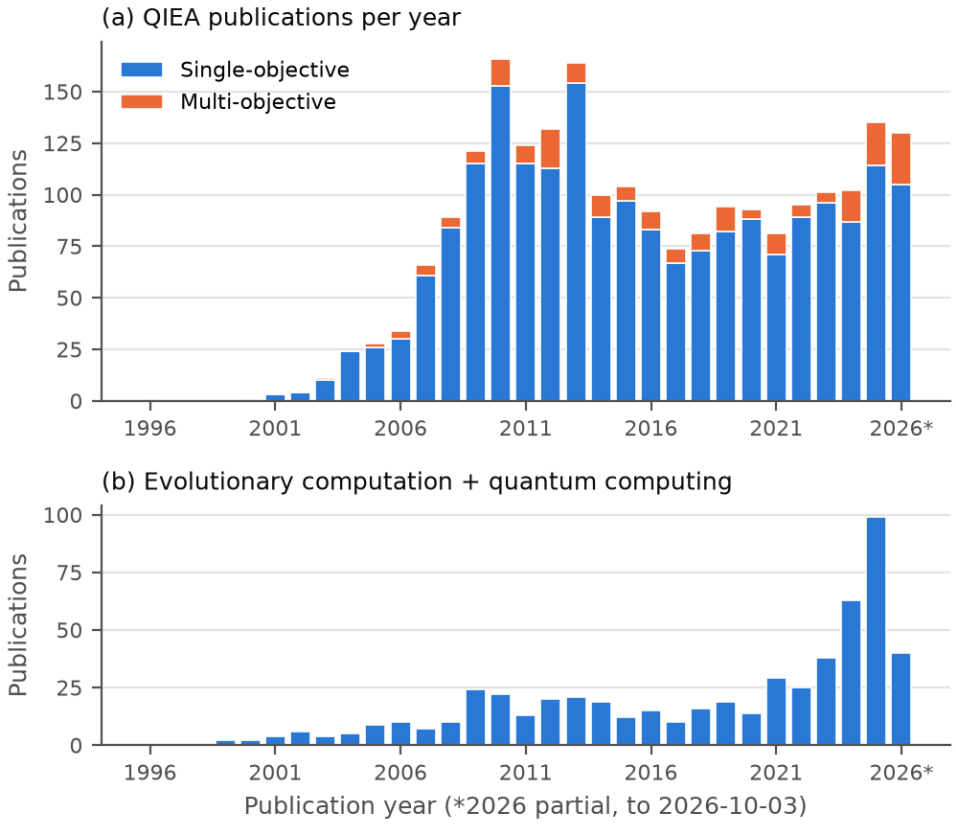


Figure 1: Publications per year, 1996–2026, from OpenAlex after cleaning (Section 2.1). (a) QIEA query (2,250 papers), split into single-objective and multi-objective papers. (b) Broad query on evolutionary computation and quantum computing (558 papers). 2026 covers papers indexed up to 3 October 2026 only.

2.3 Keyword Clusters

The keyword network splits into four clusters (Figure 2, Table 1):

Cluster 1, search dynamics and operators (20 keywords). Convergence rate, premature convergence, the quantum rotation gate, population diversity, and the exploration–exploitation balance. This is the algorithm-design core of the field, corresponding to the representation and operator axes of Sections 4.2 and 4.3.

Cluster 2, combinatorial benchmarks (9 keywords). Combinatorial optimization, the knapsack and travelling salesman problems, and superposition: the lineage of the original single-objective papers [2, 20].

Cluster 3, multi-objective engineering applications (15 keywords). Multi-objective optimization and the Pareto front, together with wireless sensor networks, power-loss reduction, distributed generation, makespan minimization, resource allocation, and energy efficiency.

Cluster 4, hybrids with other methods (11 keywords). Genetic algorithms, particle swarm optimization (including its quantum-behaved variant), feature selection, support vector machines, and hyperparameter optimization: QIEAs combined with, or compared against, other metaheuristics and machine-learning models.

The most informative feature of this structure for this survey is where multi-objective optimization sits. It does not cluster with the operator and convergence vocabulary of Cluster 1, but with application domains. In other words, the multi-objective QIEA literature is organized mainly around *what* is being optimized, not around how a quantum-inspired operator should change when there are several objectives. This is consistent with the hybridization pattern found in Section 4.5 (a QIEA operator grafted onto an unchanged classical MOEA) and with the under-specified reference solution discussed in Section 6.2. The keywords are assigned automatically by OpenAlex rather than by the authors, so this reading should be taken as indicative rather than definitive.

2.4 Most-Cited Papers

Table 2 lists the ten most-cited papers in the QIEA corpus. Three patterns stand out. First, the list is old: apart from Narayanan and Moore [20], eight of the ten papers date from 2001–2010 and only one from the last decade, which partly reflects the time citations take to accumulate. Second, it is dominated by one research group: Han, Kim, and colleagues authored three of the ten, including the top-ranked paper, which alone has more than twice the citations of any other. Third, only two of the ten are multi-objective [22,

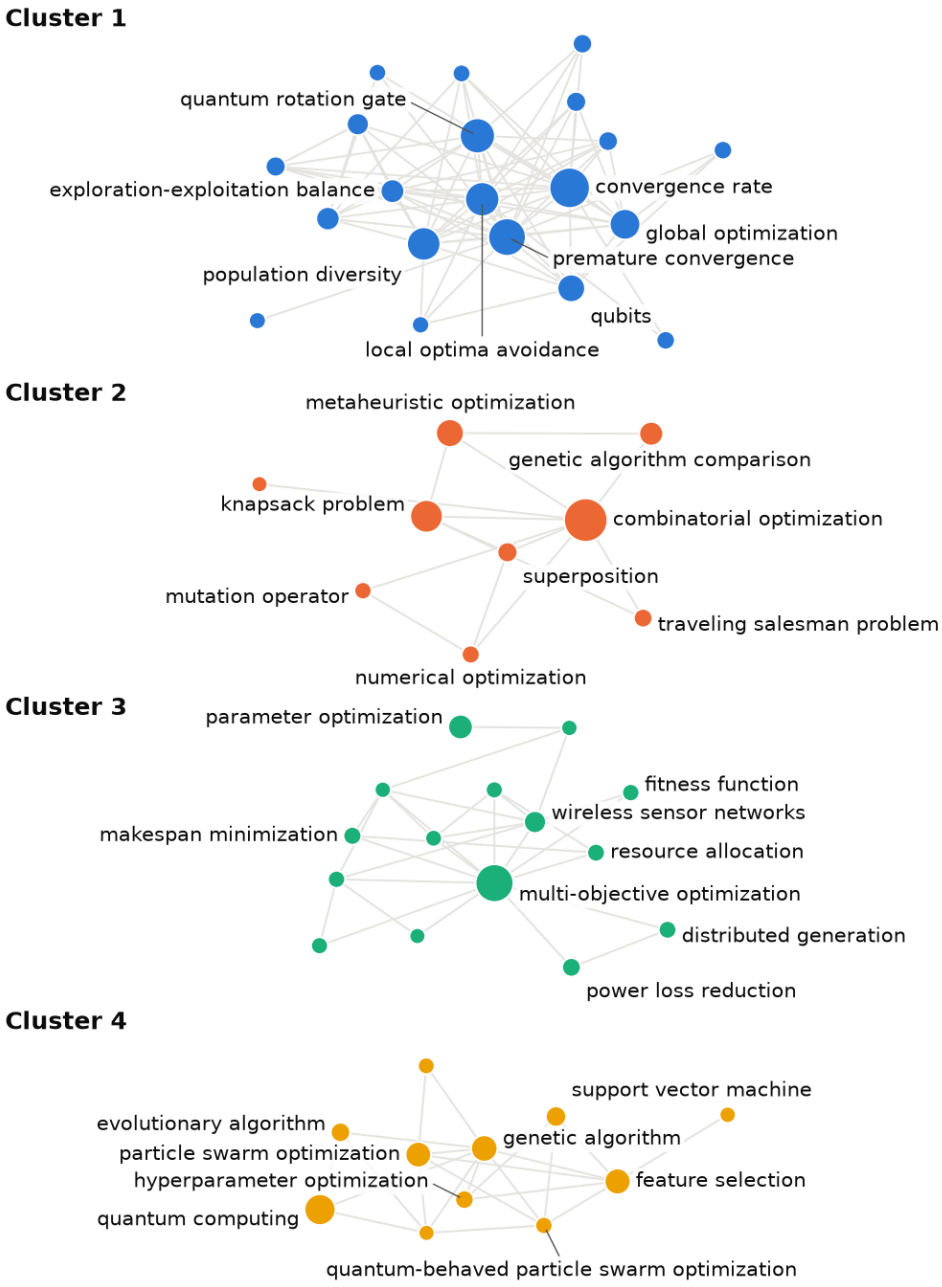


Figure 2: Keyword co-occurrence clusters in the QIEA corpus (55 keywords occurring in at least 20 papers; Louvain clustering). Node size is proportional to the number of papers with that keyword; within each cluster, the stronger half of co-occurrence links is drawn and the eight most frequent keywords are labelled. Links between clusters are not drawn. The full keyword list per cluster is in Table 1.

Table 1: Keywords in each cluster of Figure 2, most frequent first.

Cluster	Keywords
1. Search dynamics and operators	convergence rate, premature convergence, quantum rotation gate, local optima avoidance, population diversity, global optimization, qubits, exploration-exploitation balance, hybrid metaheuristic, local search, function optimization, global search, continuous optimization, global search ability, BP neural network, real-coded genetic algorithm, benchmark functions, simulated annealing, image segmentation, NP-hard combinatorial optimization
2. Combinatorial benchmarks	combinatorial optimization, knapsack problem, metaheuristic optimization, genetic algorithm comparison, superposition, traveling salesman problem, numerical optimization, mutation operator, NP-complete problems
3. Multi-objective engineering applications	multi-objective optimization, parameter optimization, wireless sensor networks, power loss reduction, makespan minimization, distributed generation, resource allocation, fitness function, Pareto front, energy efficiency, Pareto optimal solutions, quality of service, clustering, energy consumption, solution diversity
4. Hybrids with other methods	quantum computing, genetic algorithm, feature selection, particle swarm optimization, support vector machine, evolutionary algorithm, hyperparameter optimization, quantum-behaved particle swarm optimization, parameter estimation, dimensionality reduction, quantum-inspired algorithms

23], and both are application papers that hybridize a QIEA with another method, not papers about multi-objective QIEA design as such. The single-objective papers are the ones the field's representation and operators were built from, which is why Section 4.1 uses them as the baseline for the multi-objective methods. Each paper is summarized briefly below, in rank order.

Table 2: The ten most-cited papers in the QIEA corpus (OpenAlex citation counts, 3 October 2026). Year is the year OpenAlex records, which for journal papers can be the year of online publication rather than of the printed volume.

#	Paper	Contribution	Year	Citations
1	Han and Kim [2]	QEA: Q-bit chromosome and Q-gate	2002	1,589
2	Narayanan and Moore [20]	First quantum-inspired GA	1996	648
3	Han and Kim [21]	H_ϵ gate, termination criterion	2004	482
4	Deng et al. [22]	Multi-objective airport gate allocation	2020	339
5	Zhang [14]	QIEA survey and empirical study	2010	297
6	Li and Wang [23]	Multi-objective flow-shop scheduling	2007	208
7	Zhao et al. [34]	Cognitive-radio spectrum allocation	2009	205
8	Lee et al. [35]	Economic dispatch with wind power	2010	194
9	Han et al. [32]	Parallel QGA	2001	191
10	Defoin Platel et al. [36]	QEA as an estimation-of-distribution algorithm	2008	175

1. Han and Kim [2]. Introduces the quantum-inspired evolutionary algorithm (QEA): individuals are strings of Q-bits, each a probabilistic superposition of 0 and 1, and a Q-gate (rotation gate) drives them toward better solutions. On the knapsack problem, QEA performs well even with a small population and without the premature convergence of a conventional genetic algorithm. This is the representation and operator described in Section 3.1, and the one nearly every later method in this survey builds on.

2. Narayanan and Moore [20]. The first quantum-inspired genetic algorithm. Drawing on the many-universes interpretation of quantum mechanics, it runs several populations ("universes") in parallel and combines them with an interference crossover, compared against a classical genetic algorithm on the travelling salesman problem. The quantum-inspired version is shown, informally, to perform better on a small instance. It predates the Q-bit chromosome, and is the only method in this survey built around interference (Section 4.3).

3. Han and Kim [21]. Improves QEA in three ways: a termination criterion that gives a clearer meaning to the convergence of Q-bit individuals, the H_ϵ gate (a rotation gate that keeps each Q-bit probability away from exactly 0 or 1), and a two-phase scheme that tunes the Q-bits' initial conditions in a first phase. Tested on numerical and combinatorial problems, the updated QEA improves convergence speed, fitness, and robustness over the original.

4. Deng et al. [22]. A three-objective airport gate allocation model (passenger walking distance, balance of idle time across gates, and use of large gates), solved by IPOQEA, a QEA combined with a niche co-evolution strategy and an enhanced particle swarm optimizer. Validated on operational data from Baiyun Airport. The only paper of the last decade in the list, and one of its two multi-objective papers; the abstract does not state whether the three objectives are handled by Pareto ranking or combined into one.

5. Zhang [14]. A survey of QIEAs on classical computers that proposes a unified framework for them, reviews their theoretical development and their applications from combinatorial to numerical optimization, and compares the performance of different QIEA types empirically. It is the most-cited general survey of the field.

6. Li and Wang [23]. A hybrid quantum-inspired genetic algorithm for multi-objective flow-shop scheduling. A Q-bit genetic algorithm explores the binary space, with a random-key decoding that turns Q-bit strings into job permutations, while a permutation-based genetic algorithm explores and refines schedules directly. The two parts handle multiple objectives differently: the Q-bit part uses a randomly weighted sum of the objectives, while the permutation part uses non-dominated sorting with two population-trimming techniques for diversity. The random weighting is one of the few documented answers to the reference-solution question of Section 4.1: for any one weight vector there is again a single best solution to rotate toward.

7. Zhao et al. [34]. Formulates spectrum allocation in cognitive radio as an optimization problem and solves it with a genetic algorithm, a quantum genetic algorithm, and particle swarm optimization, using a mapping between the channel-assignment matrix and each algorithm's encoding to shrink the search space. All three outperform the commonly used colour-sensitive graph-colouring algorithm. The QIEA here is one of three interchangeable optimizers rather than the paper's focus, an early example of the comparative application papers that Cluster 4 captures.

8. Lee et al. [35]. Applies a quantum genetic algorithm to dynamic economic dispatch in a power system with valve-point effects and wind generation, reporting that it finds the minimum-cost dispatch quickly and accurately. Its published abstract gives no further detail of the algorithm; it is representative of the power-system applications that appear in Cluster 3.

9. Han et al. [32]. A parallel quantum-inspired genetic algorithm (PQGA). It argues that the Q-bit representation's rapid convergence and

global search make QIEAs well suited to parallel, multi-population structures, and shows on the knapsack problem that PQGA outperforms both the sequential QGA and conventional genetic algorithms.

10. Defoin Platel et al. [36]. Shows that QEA belongs to the class of estimation-of-distribution algorithms (EDAs): its Q-bit population is a set of probabilistic models of promising solutions. Compared with three classical EDAs, QEA does well on loss of diversity, scalability, solution quality, and robustness to fitness noise, which the authors attribute to its multi-model nature: it adapts its learning speed and can represent more complex distributions than a single-model EDA. This reframing matters for the multi-objective case, where a single probabilistic model has to cover a whole Pareto front rather than one optimum.

2.5 Limitations of the Analysis

Four limitations apply. OpenAlex has broad coverage but imperfect metadata: the two publication-year errors corrected above were found only because they affected highly cited papers, and others may remain in the long tail. The keywords are assigned automatically, not by authors, so the clusters describe OpenAlex's reading of each abstract. The multi-objective tag is a text match, which will miss papers that describe several objectives without these terms and include some that mention them only in passing. Citation counts favour older papers and change over time; the ranking above is a snapshot as of the retrieval date.

3 Background

3.1 Quantum Computing Concepts for QIEA

Quantum-inspired evolutionary algorithms borrow their representation and update rules from the mathematics of a single qubit, without requiring quantum hardware to run. A qubit's state is a normalized superposition of the two classical basis states,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{1}$$

where $|\alpha|^2$ and $|\beta|^2$ are interpreted as the probability of the qubit collapsing to 0 or 1 upon observation. Han and Kim [2] introduced the *Q-bit chromosome*: an individual is encoded not as a fixed binary string but as a string of such probability-amplitude pairs (α_i, β_i), one per bit position. Observing (measuring) a Q-bit chromosome collapses it into an ordinary binary solution that can be evaluated by the problem's fitness function, while the amplitudes themselves are what the algorithm evolves – maintaining, in effect, a probabilistic superposition over the whole population's search history rather than a single discrete solution.

The corresponding variation operator is the *Q-gate*, a rotation applied to each Q-bit's amplitude pair,

$$\begin{pmatrix} \alpha_i' \\ \beta_i' \end{pmatrix} = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix} \begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix}, \tag{2}$$

where the rotation angle θ_i and its sign are looked up from a predetermined table that compares the current bit's observed value and fitness contribution against the best solution found so far, nudging the amplitude pair toward whichever basis state that comparison favors. This rotation-gate mechanism is what most surveyed QIEA methods share regardless of application domain [3], and it is the representation/operator combination Section 4.2 and Section 4.3 use as the taxonomy's reference point; variants that replace or augment it with interference- or entanglement-inspired operators are discussed there rather than here.

3.2 Multi-Objective Optimization Fundamentals

A multi-objective optimization problem (MOP) seeks to simultaneously optimize $m \geq 2$ objective functions $f_1(x), \ldots, f_m(x)$ over a decision space $x \in \Omega$. Because the objectives typically conflict, there is generally no single x that minimizes (or maximizes) all of them at once; instead, a solution x_1 is said to *Pareto-dominate* x_2 if x_1 is at least as good as x_2 on every objective and strictly better on at least one. A solution is *Pareto-optimal* if no other feasible solution dominates it, and the image of the set of all Pareto-optimal solutions in objective space is the *Pareto front*. Rather than a single best answer, a multi-objective evolutionary algorithm (MOEA) therefore aims to return a population that approximates this front as closely and as evenly as possible [37].

Comparing how well two algorithms' output populations approximate the true (or best-known) Pareto front requires quality indicators that capture both convergence and diversity, since neither alone is sufficient: a tightly clustered set of points on the front converges well but covers it poorly, while a widely spread set may cover a region no solution actually dominates. The indicators most relevant to this survey – all among those commonly offered by MOEA software frameworks [38] – are:

- **Hypervolume (HV)**: the volume of objective space dominated by a solution set relative to a fixed reference point – the only known unary (single-set) indicator that is strictly monotonic with Pareto dominance [39], and the basis of hypervolume-driven selection in SMS-EMOA [8]. Its value is sensitive to the reference point and normalization used, a methodological detail Section 7.4 returns to.
- **Inverted Generational Distance (IGD)**: the average distance from each point on a reference (true or best-known) Pareto front to its near-

est neighbor in the algorithm's output set, rewarding both convergence and diversity but requiring that reference front to be known or well-approximated – a requirement this survey's own case study (Section 7) does not meet for any of its benchmark circuit families, so IGD does not itself recur there.

- **Spread and spacing**: measures of how evenly solutions are distributed along the approximated front, independent of how close that front is to the true optimum. Spread recurs in the case study (Section 7.2); spacing does not.

- **Coverage (C)**: a binary indicator that directly compares two solution sets A and B. $C(A, B)$ is the fraction of solutions in B weakly dominated by at least one solution in A [39, 40], so $C(A, B) = 1$ means every solution in B is dominated by or equal to some solution in A. Since $C(A, B)$ and $C(B, A)$ are not complementary, both directions must be reported.

- **Two Set Hypervolume Difference (TSHD)**: the hypervolume dominated by one set but not by the other, $D(A, B) = HV(A \cup B) - HV(B)$ [41]. Unlike Coverage, it measures by how much, not only whether, one set improves on the other, making it a direct hypervolume-based comparison between two algorithms' output fronts.

Unlike HV, IGD, and spread, which score a single solution set, Coverage and TSHD compare two sets directly, and so are suited to ranking algorithms against each other. Neither is computed in this survey's own case study (Section 7), which instead compares algorithms through a shared-reference hypervolume (Section 7.4). See Emmerich and Deutz [37] and Hanne and Moghaddam [42] for a fuller treatment of these fundamentals and their formal definitions.

3.3 Classical MOEAs

The Pareto-dominance-based family of MOEAs – the family every QIEA-MOO method surveyed here either builds on or is benchmarked against – traces back to Schaffer [19]'s Vector Evaluated Genetic Algorithm (VEGA), which optimized each objective in a separate sub-population and suffered from poor diversity as a result. Successive algorithms addressed this diversity problem in turn: Fonseca and Fleming [43]'s MOGA introduced Pareto-rank-based fitness with fitness sharing; Horn et al. [44]'s NPGA added dominance-based tournament selection with niching; Zitzler and Thiele [40]'s SPEA introduced an external archive of non-dominated solutions and strength-based fitness; Knowles and Corne [45]'s PAES took a different design point entirely, a (1+1) evolution strategy with no population or crossover, using an adaptive

grid archive for diversity; and Corne et al. [46, 47]'s PESA and PESA-II introduced a hyper-grid ("hyperbox") archive with selection performed at the hyperbox level. This lineage matters for QIEA-MOO specifically because several surveyed methods graft a Q-bit/Q-gate representation onto one of these selection mechanisms rather than inventing a new one – understanding the mechanism being reused is a precondition for the taxonomy's hybridization axis (Section 4.5).

Five classical MOEAs serve as this survey's background reference set. In practice NSGA-II is by far the baseline QIEA-MOO methods are most often compared against (Section 5); the other four are included because together they span the main alternative selection paradigms:

- **NSGA-II** [4]: fast non-dominated sorting (ranking the population into successive Pareto fronts) combined with crowding-distance-based diversity preservation within each front; the most widely used MOEA baseline, and the one this survey's own case-study pipeline builds on (Section 7).
- **NSGA-III** [5]: replaces crowding distance with reference-point-based niching, extending NSGA-II's non-dominated-sorting framework to many-objective problems ($m \gg 3$) where crowding distance alone provides insufficient selection pressure.
- **SPEA2** [6]: refines SPEA with a finer-grained strength-based fitness assignment, a k-th-nearest- neighbor density estimator, and a fixed-size external archive.
- **MOEA/D** [7]: decomposes the multi-objective problem into a set of scalar subproblems via weight vectors, solving them cooperatively using neighborhood information – a fundamentally different, decomposition-based design point from the three Pareto-dominance-based algorithms above.
- **SMS-EMOA** [8]: an indicator-based method whose environmental selection removes whichever individual contributes least to the population's dominated hypervolume, directly optimizing for the HV indicator described in Section 3.2 rather than a proxy for it.

Together these five span the main selection paradigms available to QIEA-MOO methods: crowding-distance-based, reference-point-based, strength/density-based, decomposition-based, and hypervolume-driven. Section 4.4 uses this same set of paradigms as one of the taxonomy's classification axes – and finds that the surveyed methods draw on the first and fourth of them, alongside archive-, grid-, and preference-based mechanisms, but never on the second, third, or fifth.

4 Taxonomy

We organize the surveyed QIEA-for-MOO literature along five axes: representation scheme, quantum-inspired operator type, selection/diversity mechanism, hybridization with classical MOEAs, and application domain. The axes were derived inductively from recurring design choices across the literature identified in Section 1.1, rather than imposed from an existing classification. They were not derived from the bibliometric sweep of Section 2, but two of its keyword clusters correspond to two of them: one gathers the operator and convergence vocabulary of the operator axis, another the application domains (Section 2.3). Section 6 revisits which surveyed methods are described in enough replicable detail to place along them with confidence. The five axes are not independent – a method's operator type constrains which representations it can pair with, and its selection mechanism is often inherited wholesale from the classical MOEA it hybridizes with – but each highlights a different design decision worth comparing across methods, and Section 5 uses them as the columns of its systematic comparison table. Before the five axes, Section 4.1 sets out the contrast between single- and multi-objective QIEAs, which cuts across all of them.

4.1 Single- vs. Multi-Objective QIEAs

Single-objective QIEAs are not the subject of this survey, but they are the baseline every QIEA-MOO method starts from, and comparing the two shows which parts of a QIEA are tied to having one objective. Most of the machinery carries over unchanged. The Q-bit chromosome, observation into a binary string, and the rotation gate of Eq. 2 are the same in Han and Kim [2]'s single-objective QEA and in Kim et al. [10]'s multi-objective QMEA, which applies them to a multi-objective knapsack problem without changing the encoding.

What does not carry over is the rotation gate's *reference solution*. The gate rotates each Q-bit toward the corresponding bit of a reference solution, and the single-objective QEA has an obvious choice: the best solution found so far, kept per individual and refreshed by local and global migration [2]. With several objectives there is no single best solution, only a set of mutually non-dominated ones, so every QIEA-MOO method must decide which of them each Q-bit individual rotates toward. The QMEA/MQEA line of work answers by updating each Q-bit individual toward non-dominated solutions in an archive [48], drawn at random, a step its authors call global random migration [49]. This turns the reference choice into a diversity mechanism in its own right: different individuals are pulled toward different parts of the front rather than all toward one point. Li and Wang [23]'s hybrid quantum-inspired genetic algorithm for multi-objective flow-shop scheduling gives a second answer: its Q-bit component scalarizes the objectives with randomly

drawn weights, so that for each weight vector there is again a single best solution to rotate toward. Deng et al. [25]'s MOQEA/D gives a third: it decomposes the problem into single-objective subproblems and assigns each Q-bit string its own, so each string again has a single best solution. Outside the thirteen methods, Ho et al. [50] go further and redesign the QEA for multiple objectives, with a fitness assignment that counts how many objectives a solution improves and by how much, and a redesigned rotation-angle update. For the NSGA-II-hybridized methods (Section 4.5), the sources gathered for this survey describe the outer NSGA-II loop in detail but do not, at the level of detail we could confirm, state how the rotation gate's reference solution is chosen.

Two smaller differences follow from the same cause. First, the single-objective QEA keeps one reference solution per individual, while a QIEA-MOO method must keep an archive of non-dominated solutions, which adds memory and a per-generation archive-update cost. Second, the single-objective literature treats Q-bit convergence as the goal: Han and Kim [21] use the convergence of Q-bit probabilities toward 0 or 1 as a termination criterion, and introduce the H_ϵ gate specifically to keep those probabilities from collapsing completely. In a multi-objective setting, a population whose Q-bits all collapse toward one binary string has lost the diversity it needs to cover a Pareto front, so a mechanism like the H_ϵ gate's lower bound arguably matters more there than in the single-objective case it was designed for. Section 6.2 returns to what these differences imply for the surveyed methods.

4.2 Representation Scheme

The representation axis asks what an individual encodes and how that encoding is turned into an evaluable candidate solution. The canonical QIEA representation is Han and Kim's Q-bit chromosome (Section 3.1, Eq. 1): a string of probability-amplitude pairs that is *observed* into an ordinary binary string before fitness evaluation, while the amplitudes themselves carry the evolutionary search's memory across generations [2]. This representation transfers directly to combinatorial 0/1 problems – Kim et al. [10]'s QMEA observes a Q-bit chromosome into the binary decision vector of a multi-objective knapsack problem essentially unchanged from the single-objective case – but requires adaptation for continuous decision spaces, where the decision vector is real-valued rather than binary: Olvera et al. [11] keep observation-as-measurement unchanged but add an explicit decoding step after it, converting each observed binary string into floating-point decision variables before fitness evaluation – so that the precision with which a real value can be represented is bounded by how many Q-bits are allotted to it. Feature subset selection sits at the combinatorial end of this spectrum and is, per Vivek et al. [3]'s survey, one of the domains where the Q-bit representation is used closest to its original binary-observation form, since a feature

is naturally either included or excluded.

Notably, this survey's own case study (Section 7) does not use a Q-bit representation at all: its individuals are classical, domain-specific chromosomes – lists of gate tuples `(gate_name, target_qubit, ctrl_qubit, angle)` – evolved with ordinary (non-quantum-inspired) genetic operators. This is not an oversight but a reminder that the representation axis is orthogonal to whether a method's *problem domain* happens to be quantum computing; we return to this distinction under application domain (Section 4.6).

4.3 Quantum-Inspired Operator Type

The operator axis asks which quantum-mechanical phenomenon a method draws its variation operator from. The dominant choice by far, across essentially every method surveyed, is the rotation-gate operator of Eq. 2, which nudges each Q-bit's amplitude pair toward whichever basis state a lookup table favors given the current individual's fitness relative to the best solution found so far [2, 3]. Beyond the rotation gate, two further quantum phenomena recur, layered on top of rather than replacing it: superposition, exploited by Güzel et al. [12]'s QNSGA-II to diversify sampling around the current population and reportedly speed convergence relative to a plain NSGA-II baseline, and entanglement, exploited by Kumar et al. [13]'s EAH-QNSGA-II as an entanglement-driven crossover operator that couples pairs of individuals more tightly than independent single-point crossover would, combined with an adaptive (rather than fixed-table) quantum rotation step and a local-search refinement pass. The word "superposition" alone does not mark a method as quantum-inspired, however: Florea et al. [51]'s "superposition of populations" runs NSGA-II and SPEA2 concurrently, each producing a share of every generation set by its recent solution quality, with no quantum-inspired representation or operator involved. Interference – the third quantum phenomenon named alongside superposition and entanglement in Section 1 as a general source of QIEA design inspiration – has no multi-objective method in the currently surveyed set built explicitly around it as a variation operator. It does appear on the single-objective side, and earlier than the rotation gate itself: Narayanan and Moore [20]'s quantum-inspired genetic algorithm for the travelling salesman problem, the earliest QIEA we cover, keeps several parallel "universes" of solutions and builds new individuals with an *interference crossover* that takes successive genes from successive universes. Interference was therefore part of quantum-inspired evolutionary design from the start, but did not survive into the rotation-gate-based multi-objective methods surveyed here. Section 6.1 considers what this narrow operator range implies for the field.

4.4 Selection and Diversity Mechanism

The selection axis asks how a method ranks and prunes its population across objectives – the same paradigms introduced as classical MOEA baselines in Section 3.3: crowding-distance-based, reference-point-based, strength/density-based, decomposition-based, and hypervolume-driven. Across the thirteen surveyed methods, selection is more varied than the recent literature alone suggests. The three recent NSGA-II-based methods inherit NSGA-II's selection *unchanged*: Güzel et al. [12]'s QNSGA-II and Kumar et al. [13]'s EAH-QNSGA-II both retain its crowding distance verbatim, as do the three continuous-space algorithms of Olvera et al. [11], applying their quantum-inspired operators only to how new candidates are generated, not to how the population is ranked or pruned. The older QMEA/MQEA line ranks by Pareto dominance and keeps an external non-dominated archive [10, 48], and MQEA-PS fills that archive by user preference instead of dominance [24]. Zhang et al. [29] maintain diversity with a self-adaptive grid, in the spirit of PAES and PESA [45, 46]. Deng et al. [25]'s MOQEA/D decomposes the problem into single-objective subproblems, the MOEA/D paradigm. Li and Wang [23] combine a randomly weighted sum of the objectives, in their Q-bit component, with non-dominated sorting and population trimming in a permutation-based component. The remaining four methods do not state their selection mechanism in their abstracts [22, 26–28]. What none of the thirteen does is pair a Q-bit representation with reference-point-based (NSGA-III-style), strength/density-based (SPEA2-style), or hypervolume-driven (SMS-EMOA-style) selection, even though the first two have recently gained the strongest theoretical guarantees of their lineages (Section 6.1). Section 6 treats this absence as a candidate limitation rather than evidence that these paradigms are unsuitable. This project's own case study pairs all three of NSGA-II, NSGA-III, and SMS-EMOA selection with a classical (non-quantum-inspired) chromosome, so it demonstrates the reverse pairing is viable, but on the domain- specific representation side of the representation axis (Section 4.2), not the Q-bit side – it does not by itself close this gap for QIEA-MOO specifically.

4.5 Hybridization with Classical MOEAs

Closely related to the selection axis, the hybridization axis asks how tightly a method's quantum-inspired representation and operators are grafted onto an existing classical MOEA's overall algorithmic skeleton, versus how much of the surrounding framework is bespoke. On this axis, the surveyed methods fall into three design points. In the first, which the three most recent methods share, a method names and frames itself as an explicit variant of a single classical MOEA – NSGA-II in every case found – with the Q-bit representation and one or two quantum-inspired operators substituted into

the generation step, while the outer non-dominated-sorting-plus-crowding-distance loop is left otherwise intact (Güzel et al. [12], Kumar et al. [13]). Olvera et al. [11] follow the same pattern twice over: their proposed SQMEA is explicitly a quantum-inspired version of NSGA-II, and NSGA-II (in both binary- and floating-point-coded form) is also one of the classical baselines they compare against. A second design point, older and more common among the most-cited methods, extends a QIEA with its own multi-objective machinery rather than grafting it onto a named MOEA: Kim et al. [10]'s QMEA and its successors [24, 48] build Pareto ranking and an archive directly into the QEA, Zhang et al. [29] add an adaptive grid, and Deng et al. [25] borrow MOEA/D's decomposition idea without presenting the result as a variant of MOEA/D itself. A third design point hybridizes the QIEA with another metaheuristic rather than with an MOEA: Li and Wang [23] pair a Q-bit genetic algorithm with a permutation-based genetic algorithm, the first exploring a binary space and the second refining schedules directly, and Deng et al. [22] combine a QEA with particle swarm optimization and a niche coevolution strategy. These two are also the most-cited multi-objective QIEAs in the bibliometric corpus, so the NSGA-II-variant design point, although the most common among recent methods, is also the newest of the three and so far the least cited. This survey's own case study (Section 7) sits at the opposite end of the same axis from the QIEA-MOO literature above: it hybridizes classical MOEAs (NSGA-II, NSGA-III, SMS-EMOA, used as unmodified per-block optimizers) with a domain-specific, non-quantum-inspired chromosome, rather than hybridizing a classical MOEA with a Q-bit representation – the same "reverse pairing" noted under selection above, presented here as a structural contrast rather than as an example of QIEA-MOO hybridization itself.

4.6 Application Domain

The application-domain axis asks what kind of problem a method is evaluated on. Two largely disjoint clusters emerge across the literature surveyed here. The first cluster comprises classical combinatorial or engineering optimization problems to which a quantum-*inspired* algorithm is applied on ordinary classical hardware: multi-objective 0/1 knapsack [10, 30], continuous and DTLZ benchmark problems [11, 24], flow-shop and cloud-workflow scheduling [23, 26], vehicle routing [29], airport gate allocation [22, 25], robot path planning [24], UAV task allocation [27], electric-vehicle powertrain design [28], and electric-vehicle charging-station siting [13]. In every one of these, "quantum" describes the algorithm's inspiration, not the problem: knapsacks, schedules, routes, airport gates, vehicles, and charging networks are entirely classical objects. The second cluster is the opposite pairing: quantum circuit optimization, where the problem domain is genuinely quantum (a target unitary or circuit to reproduce) but the optimizer applied to

it, in this survey's own case study and its foundational reference [18], is an unmodified classical MOEA rather than a quantum-inspired one. The two do meet on the single-objective side: the bibliometric corpus of Section 2 contains QIEAs that evolve quantum circuits at the gate level [52], design or synthesize circuits for given gate libraries [53, 54], and search parameterized-circuit architectures for quantum neural networks [55]. We did not, however, find a *multi-objective* QIEA applied to a quantum-computing problem domain, even though such problems are naturally multi-objective – fidelity against depth and gate count in this survey's own case study. Whether that combination would offer any advantage over either cluster alone is an open question we return to as a candidate future direction in Section 8.

5 Feature Comparison

Section 4 introduced five classification axes and applied them narratively to the QIEA-for-MOO methods surveyed. This section applies the same axes systematically, one row per method, across two tables: Table 3 recaps representation, operator type, selection mechanism, and application domain, and Table 4 turns to complexity, scalability, convergence behavior, benchmarks, and reported performance – the remaining dimensions RQ2 (Section 1.2) asks about. A third, smaller table (Table 5) lists the single-objective QIEAs included as a comparison baseline (Section 4.1).

A methodological caveat applies to Table 4 specifically, and is itself a preview of Section 6's reproducibility discussion: its cells report what each paper itself states, not an independent re-derivation or reproduction of its results. Where a paper does not state a quantity clearly enough to cite with confidence, the cell is marked "not confirmed" rather than filled with an unverified number. That so many cells end up marked this way, in what is meant to be the survey's most concrete deliverable, is itself evidence for a pattern Section 6 treats as a cross-cutting limitation: QIEA-MOO papers routinely under-report the complexity, scale, and statistical rigor of their own evaluations relative to what a reader would need to compare them fairly against one another.

The application-domain column of Table 3 is worth pausing on before the tables themselves: every method in it targets a classical combinatorial, continuous, or engineering problem – a knapsack, a benchmark function, a schedule, a charging network – and none targets a quantum circuit. Figure 3 shows, for contrast, a small instance of exactly that other kind of object: a 4-qubit quantum Fourier transform circuit, the sort of target this survey's own case study (Section 7) evolves a cheaper replacement for. No method surveyed in this section operates on an object like it – a gap Section 4.6 already named and Section 6.1 returns to.

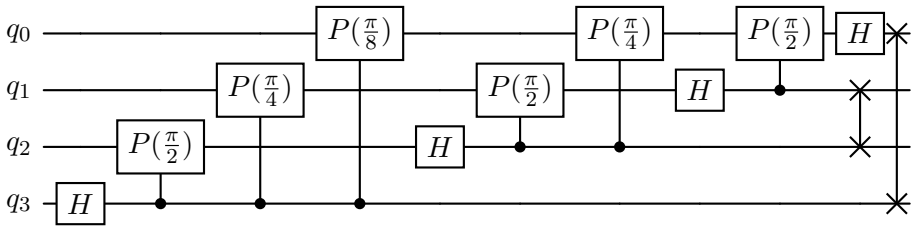


Figure 3: A 4-qubit quantum Fourier transform circuit (H, controlled phase rotations, and a final `SWAP` layer), gate for gate as produced by this survey's own case-study pipeline's circuit generator (Section 7.1). Shown here purely for contrast with the table below: none of the QIEA-MOO methods surveyed in this section target an object like it.

Table 3: Representation, operator, selection, and domain summary of the thirteen surveyed QIEA-MOO methods, in order of publication (recap of the taxonomy axes, Section 4; method selection rule in Section 1.1).

Method	Representation	Quantum-Inspired Operator(s)	Selection / Diversity	Application Domain
QMEA [10]	Q-bit chromosome, observed to a binary decision vector	Rotation gate toward archive solutions drawn at random [48, 49]	Pareto-dominance ranking with an external non-dominated archive (no crowding distance)	Multi-objective 0/1 knapsack (combinatorial)
HQGA [23]	Q-bit chromosome, observed to binary, then random-key decoded to a job permutation; paired with a permutation-coded GA	Rotation gate + Q-bit genetic operators	Randomly weighted sum of objectives (Q-bit part); non-dominated sorting with two population-trimming techniques (permutation part)	Multi-objective flow-shop scheduling (combinatorial)
MQEA-PS [24]	Q-bit chromosome (as in MQEA)	Rotation gate toward archive solutions	Non-dominated sorting of subpopulations; archive filled by preference-based selection (fuzzy-integral scoring of user preferences)	DTLZ benchmarks; fuzzy path planner for a soccer robot
MOQEA [29]	Not confirmed beyond the abstract	Not confirmed beyond the abstract	Non-dominated set built by "Challenge Cup" rules; self-adaptive grid for diversity	Multi-objective vehicle routing with customer satisfaction
APMQEA [30]	Not confirmed (no abstract available)	Not confirmed	Adaptive population size; other details not confirmed	Multi-objective 0/1 knapsack (combinatorial)

Continued on next page

Table 3 (continued)

Method	Representation	Quantum-Inspired Operator(s)	Selection / Diversity	Application Domain
QGA for powertrain design [28]	Not confirmed	Quantum genetic algorithm (details not in the abstract)	Not stated (vehicle dynamics and economy as the two indexes)	Dual-motor electric-vehicle powertrain matching
QGA for UAV coalitions [27]	Not confirmed	QEA-inspired leader–follower coalition formation	Not stated (three objectives: resource use, reliability, trust)	Task allocation in large-scale UAV networks
IPOQEA [22]	QEA individual (encoding details not confirmed beyond the abstract)	QEA update, hybridized with enhanced PSO and niche co-evolution	Not stated in the abstract (Pareto ranking vs. scalarization not confirmed)	Three-objective airport gate allocation (real-world operations)
MOQIGA [26]	Not confirmed (no abstract available)	Not confirmed	Not confirmed	Healthcare workflow scheduling in hybrid clouds with hard and soft deadlines
QNSGA-II [12]	Q-bit chromosome layered onto an NSGA-II individual	Rotation gate + superposition-based sampling	NSGA-II crowding distance, unmodified	General-purpose multi-objective benchmarks (not itemized in abstract-level notes)
Continuous QIEA-MOO [11]	Q-bit chromosome observed to binary, then decoded to floating-point decision variables	Rotation gate (lookup-table direction, dynamic angle magnitude)	NSGA-II-style fast non-dominated sorting + crowding distance	16 standard bi-objective continuous benchmarks (e.g. ZDT, LZ09), 4 of them constrained
MOQEA/D [25]	Q-bit strings, each solving one single-objective subproblem	Observation guided by an "optimal crossover" that preserves good genes	Decomposition into single-objective subproblems (MOEA/D-style)	Multi-objective knapsack; large-scale airport gate assignment
EAH-QNSGA-II [13]	Q-bit chromosome layered onto an NSGA-II individual	Adaptive rotation gate + entanglement-driven crossover + post-hoc local search	NSGA-II crowding distance, unmodified	Electric-vehicle charging-station siting (real-world infrastructure)

Table 4: Complexity, scalability, convergence, and reported performance of the surveyed QIEA-MOO methods. Cells marked "not confirmed" require reading the full paper, not just its abstract, before any specific claim is cited (see the methodological caveat above).

Method	Complexity	Scalability	Convergence	Reported Performance
QMEA [10]	Not confirmed; direct Pareto ranking without NSGA-II's fast non-dominated sort suggests a per-generation cost at or above NSGA-II's, not below it	Evaluated only on the knapsack instance sizes used in the original paper; not confirmed	Not confirmed	Not confirmed beyond the paper's own framing as an early QIEA-MOO proof of concept

Continued on next page

Table 4 (continued)

Method	Complexity	Scalability	Convergence	Reported Performance
HQGA [23]	Not confirmed; runs two GAs side by side, so per-generation cost is at least that of non-dominated sorting in the permutation part	Tested on several multi-objective flow-shop instances; sizes not confirmed	Not confirmed	Reported effective on several performance metrics against the compared algorithms (self-reported; baselines not confirmed)
MQEA-PS [24]	Not confirmed	DTLZ benchmarks and one robot path-planning task	Not confirmed	Better than MQEA with dominance-based selection, NSGA-II, and MOPBIL on DTLZ and path planning (self-reported)
MOQEA [29]	Not confirmed	Classical benchmark instances; sizes not confirmed	Not confirmed	Reported effective (self-reported; baselines not confirmed)
APMQEA [30]	Not confirmed	Not confirmed	Not confirmed	Solutions close to the Pareto front with a good spread (secondary summary only)
QGA for powertrain design [28]	Not confirmed	One vehicle design	Good global search and convergence (self-reported)	15.6% lower energy use than a conventional single-speed electric vehicle (a design outcome, not an algorithm comparison)
QGA for UAV coalitions [27]	Not confirmed	Simulations with large numbers of UAVs	Not confirmed	Better than the compared models across scenarios (self-reported)
IPOQEA [22]	Not confirmed	Evaluated on one airport's operational data (Baiyun Airport); scaling beyond that instance not confirmed	Not confirmed	Reports better optimization ability than the compared methods on gate allocation (self-reported; baselines not confirmed)
MOQIGA [26]	Not confirmed	Not confirmed	Not confirmed	Not confirmed
QNSGA-II [12]	Inherits NSGA-II's per-generation non-dominated-sorting cost, since selection is left unmodified (Section 4.4); the superposition-sampling step adds low-order per-Q-bit overhead	Not confirmed	Reported to converge faster than a plain NSGA-II baseline; the magnitude and statistical significance of that claim are not independently confirmed here	Improved convergence speed relative to NSGA-II (self-reported)

Continued on next page

Table 4 (continued)

Method	Complexity	Scalability	Convergence	Reported Performance
Continuous QIEA-MOO [11]	Inherits NSGA-II's non-dominated-sorting cost; the proposed SQMEA uses a single quantum register, requiring fewer qubits than the other two variants	Performance of the quantum-inspired variants degrades as the number of decision variables (and hence Q-bits) grows, attributed by the authors to the representation's intrinsic uncertainty; bi-objective only	Floating-point NSGA-II converged faster overall; quantum-inspired variants converged more slowly on ZDT3	Three QIEA-MOO variants compared against each other and against NSGA-II and SMPSO (30 runs, t-tests): no statistically significant difference from NSGA-II; SQMEA best of the three QIEA variants on constrained problems and complicated Pareto sets
MOQEA/D [25]	Not confirmed	"Large-scale" knapsack and gate-assignment instances; sizes not confirmed	Not confirmed	Effective on large-scale knapsack; good gate assignments (self-reported)
EAH-QNSGA-II [13]	Inherits NSGA-II's per-generation cost plus entanglement-crossover and local-search overhead; the local-search pass is structurally analogous to the local-search refinement in this survey's own case study (Section 7.3), applied after the main GA loop rather than during it	Evaluated on a single real-world EV-charging-siting case study; scaling behavior beyond that instance not confirmed	Not confirmed	Reports improved siting trade-offs on its case study relative to baseline planning approaches; specific metrics not confirmed here

Read across both tables, three patterns stand out even before the critical analysis proper (Section 6). First, selection mechanisms are more varied than the recent NSGA-II-based methods alone would suggest: the thirteen methods use NSGA-II's crowding distance [11–13], Pareto ranking with an external archive [10], preference-based archive selection [24], an adaptive grid [29], decomposition [25], and weighted sums [23]. What is absent is equally clear: none uses reference-point (NSGA-III-style), strength-based (SPEA2-style), or hypervolume-driven (SMS-EMOA-style) selection. Second, four of the thirteen, all among the most cited, do not say in their abstracts how their several objectives are combined at all [22, 26–28] – whether by Pareto ranking or by collapsing them into one weighted objective, which would make the method single-objective in practice. Third, reported performance is predominantly self-comparative – a proposed method against a classical baseline, usually NSGA-II – rather than cross-compared against other QIEA-MOO methods: in the sources gathered for this survey, the only

such comparisons are against the authors' own variants, as when Kim et al. [24] compare MQEA-PS with its predecessor MQEA. Olvera et al. [11] are an instructive partial exception: they compare three QIEA-MOO variants against each other as well as against NSGA-II and SMPSO, over 16 standard benchmark problems, 30 runs each, with statistical hypothesis testing – and it is precisely this more careful protocol that surfaces a limitation (degrading performance as the number of decision variables grows) that single-instance evaluations elsewhere in the table are not positioned to detect. These observations recur as cross-cutting limitations in Section 6.3.

Table 5 sets the single-objective baseline beside these two tables. Its last column replaces the selection mechanism with the rotation gate's reference solution, the component Section 4.1 identifies as the main thing that changes between the two settings. Read against Table 3, it shows that the representation and operator columns barely move between single- and multi-objective QIEAs (Q-bit chromosome plus rotation gate on both sides, with Narayanan and Moore [20]'s pre-Q-bit interference crossover the one exception), while the reference-solution question has a single obvious answer on the single-objective side and, on the multi-objective side, only three documented answers among the thirteen methods: random draws from a non-dominated archive in the QMEA/MQEA line [48, 49], random objective weights in Li and Wang [23], and one single-objective subproblem per Q-bit string in Deng et al. [25].

The table also shows, on the single-objective side, the scaling limitation that Section 6.3 raises for the multi-objective one. Gîlea et al. [56] – this survey's first author's own earlier work, evaluated on a single-objective formulation even though the problem it targets is naturally multi-objective – found its QIEA ahead of a classical genetic algorithm on small TSPLIB instances but behind it as the number of cities grew, and behind it on all three real routes of 199–334 locations (routes and genetic-algorithm baseline from Banciu et al. [57]), at about 3.85 times the running time. The authors attribute the loss to the repair step: as the number of cities grows, more of each tour is produced by repairing duplicates than by observing Q-bits, and the population loses diversity. This mirrors Olvera et al. [11]'s finding for continuous problems, where the quantum-inspired variants also lost ground as the number of decision variables grew: in both cases the cost of turning Q-bits into a valid solution grows with problem size.

6 Critical Analysis and Limitations

6.1 Per-Category Weaknesses

This subsection revisits each of the five taxonomy axes from Section 4 in turn, this time asking not how the surveyed methods differ along that axis but what limitation the axis itself exposes once the methods are compared

Table 5: Single-objective QIEAs included as a comparison baseline (Section 4.1). The last column gives the solution the variation operator steers toward – the component that has no direct multi-objective equivalent.

Method	Representation	Quantum-Inspired Operator(s)	Reference Solution	Application Domain
QIGA [20]	Classical permutation strings, held in several parallel "universes" (no Q-bit chromosome)	Interference crossover (successive genes taken from successive universes)	None in the rotation-gate sense; recombination across universes	Travelling salesman problem (combinatorial)
QEA [2]	Q-bit chromosome, observed to a binary string	Rotation gate (lookup-table direction and magnitude)	Best solution found so far, kept per individual and refreshed by local and global migration	0/1 knapsack (combinatorial)
QEA with H_ϵ gate [21]	Q-bit chromosome, observed to a binary string	H_ϵ gate: rotation gate with a lower bound ϵ on each Q-bit probability, plus a two-phase scheme	Best solution found so far, as in QEA	Numerical and combinatorial optimization problems (abstract level)
QIEA for waste-collection routing [56]	Q-bit chromosome; each tour position is encoded by $\lceil \log_2 N \rceil$ Q-bits, observed to a city index, and the resulting sequence is repaired into a valid permutation	Rotation gate with a decaying angle, Pauli-X quantum mutation, and qubit-level crossover	Best individual of the current generation	Travelling salesman problem: TSPLIB instances (14–52 cities) and three real waste-collection routes (199–334 locations)

against each other.

Representation. The Q-bit chromosome transfers cleanly to combinatorial 0/1 problems [10] but has no single standard decoding convention for continuous decision spaces: where a binary problem simply observes each Q-bit, Olvera et al. [11] must additionally decode each observed binary string into floating-point values, fixing how many Q-bits represent each variable – a design decision with no equivalent in the original formalism [2] to constrain it, and one with a measurable cost: the same study found the quantum-inspired variants' performance degrading as the number of decision variables, and so of Q-bits, grew. Two papers that both describe themselves as using "the Q-bit representation" may therefore be doing meaningfully different things once a continuous problem is involved, which weakens how much a shared representation label actually tells a reader trying to compare methods across domains. Compounding this, Vivek et al. [3]'s own survey of QIEA operators finds the representation used almost exclusively on combinatorial or binary problems (feature subset selection foremost among them); the continuous adaptation remains comparatively rare and non-standardized, which limits how confidently "the Q-bit representation" can be described as a general-purpose MOO representation rather than one best suited to combinatorial problems specifically.

Quantum-inspired operator type. Section 4.3 already noted that interference, named alongside superposition and entanglement as a general source of QIEA design inspiration (Section 1), has no dedicated implementation in the multi-objective methods surveyed here, even though the earliest single-objective QIEA was built around an interference crossover [20]. Taken together with how much of each surveyed method's actual variation step reduces to the single rotation-gate update rule from Han and Kim [2] – superposition-based sampling in Güzel et al. [12] and entanglement-driven crossover in Kumar et al. [13] are each one additional operator layered on top of, not a replacement for, that same rotation gate – this raises a concrete risk for the field: that "quantum-inspired" is doing more rhetorical work in motivating a paper's framing than algorithmic work in its actual contribution, since the operator-type axis in practice offers much less diversity than the three-phenomenon (superposition, interference, entanglement) framing common in this literature's introductions would suggest.

Selection and diversity mechanism. The thirteen surveyed methods use a range of selection mechanisms – NSGA-II's crowding distance, Pareto ranking with an archive, preference-based archive selection, an adaptive grid, decomposition, and weighted sums (Section 4.4) – but two weaknesses stand out. First, four of the thirteen, all among the most cited, do not state in their abstracts how their objectives are combined at all [22, 26–28]. A reader cannot tell from the abstract whether such a method searches for a Pareto front or optimizes a single weighted objective, which is the most basic fact about a multi-objective method. Second, none of the thirteen pairs

the Q-bit representation with NSGA-III's reference-point niching, SPEA2's density estimator, or SMS-EMOA's hypervolume-driven selection. This gap is worth weighing against results that postdate most of the surveyed QIEA-MOO papers: Deng et al. [58] give NSGA-III the first proven approximation guarantees of its lineage, and Alghouass et al. [59] prove SPEA2 achieves a polynomial-time-optimal approximation of the Pareto front on a benchmark problem where NSGA-II's best proven guarantee is only roughly a factor of two. Neither result implies NSGA-II's crowding-distance selection is a poor empirical choice in general, and neither has been evaluated in combination with a Q-bit representation specifically – but they do mean the QIEA-MOO literature surveyed here has not yet adopted the two selection paradigms that now have the strongest theoretical grounding, while its most recent methods have converged on NSGA-II's.

Hybridization with classical MOEAs. Hybridization in the three NSGA-II-based methods is uniformly shallow in the same direction: a Q-bit representation and one or two quantum-inspired operators are grafted onto an otherwise unmodified classical MOEA loop (Section 4.5), rather than the classical algorithm's selection or archiving logic itself being redesigned around what a quantum-inspired representation makes available (for instance, using the amplitude pair's own uncertainty as a diversity signal, rather than only its observed bit value). The older QMEA/MQEA line goes further, building its archive into the QEA itself [24, 48], and Ho et al. [50], outside the thirteen methods, redesign the QEA's fitness assignment and rotation-angle update for several objectives – so a deeper integration has been shown to be possible, but the recent methods have not followed it. A second, related weakness is evaluative rather than architectural: as already noted from Table 4 (Section 5), every reported performance claim in the surveyed set that we could confirm compares the proposed method against a classical baseline – NSGA-II in most cases, conventional planning approaches in Kumar et al. [13]'s case – and no two independently published QIEA-MOO methods among the thirteen are benchmarked against each other in the sources gathered for this survey. The only QIEA-MOO-to-QIEA-MOO comparisons are against the authors' own variants: Kim et al. [24] against their earlier MQEA, and Olvera et al. [11] across three variants they adapted themselves. A field in which new methods are routinely shown to beat the same baseline, but rarely compared against their own siblings, cannot yet support a claim about which hybridization choice is preferable within the QIEA-MOO cluster itself.

Application domain. The application domains covered split into two disjoint clusters with no overlap (Section 4.6): quantum-inspired algorithms applied to classical combinatorial or engineering problems, and classical (non-quantum-inspired) algorithms applied to a genuinely quantum problem domain, as in this survey's own case study (Section 7) and its foundational reference [18]. Within the first cluster, most papers additionally validate on a

single problem or a single real-world instance – one EV-charging-siting case study, one airport's operational data [22], one vehicle design [28] – rather than a shared benchmark suite spanning multiple problem sizes or domains; the continuous-space study of Olvera et al. [11], with its 16-problem standard suite, and MQEA-PS's use of the DTLZ suite [24] are the exceptions. Claims of general applicability therefore rest on a narrower evidentiary base than the breadth of domains represented across the cluster as a whole might suggest.

6.2 From Single to Multiple Objectives

Comparing the surveyed QIEA-MOO methods with their single-objective baseline (Section 4.1, Table 5) exposes a limitation that none of the five axes above shows on its own: the move to several objectives was made mostly by *adding* classical MOO machinery around an unchanged single-objective QIEA core, not by rethinking that core; Ho et al. [50]'s redesigned fitness assignment and angle update are the clearest exception we found.

The rotation gate's reference solution is under-specified. In the single-objective QEA, the rotation gate always has a well-defined target, the best solution found so far [2]. A multi-objective method has to replace that target with something chosen from a set of non-dominated solutions, and this choice decides where on the front each Q-bit individual is pulled. It is therefore as much a part of the method's diversity mechanism as its crowding distance or archive. Yet among the thirteen surveyed methods, only three answers are documented – a random draw from a non-dominated archive in the QMEA/MQEA line [48, 49], a randomly weighted sum of the objectives in Li and Wang [23]'s hybrid, and one single-objective subproblem per Q-bit string in Deng et al. [25]'s MOQEA/D. The only comparison between alternatives is within the MQEA line, where Kim et al. [24] change which solutions enter the archive the references are drawn from (preference-based instead of dominance-based) and report better results. No method compares choosing the reference by crowding distance, by reference direction, or by hypervolume contribution. The selection-axis finding above (four of the most-cited methods do not state how they combine objectives) has a counterpart here: the part of the algorithm that is genuinely new in the multi-objective setting is also the part least often specified or studied.

Convergence means different things in the two settings. Single-objective QIEA design treats Q-bit probabilities converging to 0 or 1 as the desired end state, and uses it as a termination criterion [21]. In a multi-objective setting, that same convergence means the population is collapsing onto one point of a front it is supposed to cover. Mechanisms that slow or bound it, such as the H_ϵ gate's floor on each probability [21], were designed for the single-objective case. We did not find a surveyed QIEA-MOO method that adapts them to, or evaluates them for, the multi-objective case.

Single-objective benchmarks transfer more directly than single-objective results. The 0/1 knapsack problem is used on both sides, by Han and Kim [2] and in multi-objective form by Kim et al. [10], which makes it the one place where single- and multi-objective QIEA results could in principle be compared directly. Even there, the comparison is indirect: a single-objective QEA's advantage over a classical GA does not imply anything about a QIEA-MOO method's advantage over NSGA-II, since the component that differs between them (the reference solution, above) is exactly the one the single-objective result says nothing about. Broad single-objective comparisons such as Zhang [14]'s empirical study of QIEA variants have no multi-objective counterpart among the sources gathered for this survey.

6.3 Cross-Cutting Open Problems

Three problems recur across every taxonomy category above rather than being specific to one of them.

Reproducibility gaps. Building Table 4 (Section 5) directly surfaced this problem: several of its complexity, scalability, and convergence cells could not be filled with confidence from each paper's title and abstract alone, and were marked accordingly rather than guessed at. This is not merely a limitation of how this survey gathered its literature – it reflects that the information needed to independently reproduce or even carefully re-derive a QIEA-MOO paper's headline result is often not recoverable without the full paper and, in many cases, code or data that may never have been released. This gap is not specific to QIEA-MOO: Pătrăuşanu et al. [38]'s systematic review of MOEA software frameworks (PlatEMO, pymoo, jMetal, Evolver, and FADSE) notes that authors very rarely make their algorithms' implementations public, and argues for common, extensible platforms that would both expose those implementation details and give the field a shared basis for benchmarking. This survey's own case study offers a cautionary illustration of how easily such gaps can hide even inside a codebase under active, careful maintenance: two real bugs in its inter-block gate injection stage silently distorted every quantitative fidelity result reported over several weeks of work before they were caught and fixed (Section 7.1). If a reproducibility gap of that kind can persist for weeks inside one actively-maintained pipeline with its own test suite, the equivalent risk across independently authored papers with no shared code and no obligation to re-verify a predecessor's numbers before building on them is considerably harder to rule out, not easier.

Weak benchmarking practices. The self-comparative-only pattern already noted under hybridization above – methods benchmarked against a classical baseline, usually NSGA-II, but not against a sibling QIEA-MOO method – is itself an instance of a broader benchmarking weakness: comparisons in this literature are typically run against one baseline, on one problem or instance, without the kind of multi-seed statistical treatment that would

distinguish a genuine improvement from run-to-run variance. Olvera et al. [11]'s 30-run, hypothesis-tested comparison on a standard benchmark suite shows the more rigorous alternative is feasible within this literature. Even that, however, would not by itself close the gap: Pătrăuşanu et al. [60] observe that ZDT, DTLZ, WFG, and CEC all exhibit only smooth variation between a solution's inputs and its objective values, and propose discrete multidimensional Thomae-like functions as adversarial, deliberately rugged benchmarks in which small input changes can produce large output jumps. They further caution that such landscapes are likely to yield fragmented Pareto fronts that standard quality indicators such as hypervolume may fail to characterize adequately – a benchmark-design gap that applies to the classical baselines and QIEA-MOO methods alike. This survey's own case study underscores why that treatment matters: an early quantitative finding in the project's own results, later retracted, turned out to be a measurement artifact rather than a real algorithmic effect once re-measured carefully (Section 7.2) – exactly the kind of error a single-run, single-baseline comparison is poorly positioned to catch. Compounding this, the classical baselines this literature compares against have themselves only recently acquired the kind of formal scrutiny that would let such comparisons be interpreted rigorously: Opris et al. [61] and Deng et al. [58]'s runtime and approximation-guarantee analyses of NSGA-III, and Alghouass et al. [59]'s analogous result for SPEA2, all postdate nearly every QIEA-MOO paper surveyed here, so those papers' original empirical comparisons against NSGA-II could not have accounted for guarantees that did not yet exist.

Scalability claims rarely stress-tested. The real-world application examples in Section 6.1's application-domain discussion are each evaluated at a single scale rather than across a size sweep, so a method's reported performance says little about how it would behave on a substantially larger instance of the same problem. Where problem size does vary, the signal is not reassuring: Olvera et al. [11]'s quantum-inspired variants lost ground as the number of decision variables increased, which the authors attribute to the Q-bit representation's intrinsic uncertainty growing with the number of Q-bits. This survey's own case study again illustrates the point concretely, on the quantum-circuit-optimization side rather than the QIEA-MOO side: its own fidelity-estimation subsystem required distinct, per-circuit-family scaling fixes and qubit-count ceilings before it could be trusted at all beyond small instances (Section 7.1), a cost that a single-scale evaluation would never have surfaced. Sweeping across scale within the case study's own 225-run core baseline (Section 7.2) surfaces a second, sharper instance of the same point: mean `fidelity_final` *declines* with qubit count for four of the five circuit families tested (e.g. `qft`: 0.070 at n=8, 0.015 at n=12, 0.0005 at n=20; `hw_efficient_ansatz`: 0.098, 0.020, 0.0003 over the same sizes), while the fifth, `weak_random`, instead *improves* with qubit count (0.225, 0.362, 0.375), as Figure 4 shows directly. A single-scale evaluation of any one of these

families – exactly the evaluative pattern this subsection has already flagged as the norm in the QIEA-MOO literature surveyed above – would report only one point on what is evidently not even a consistent trend across problem instances, let alone across algorithms. We do not attempt to adjudicate here whether the declining trend reflects a genuine scaling limitation of the pipeline or simply the exact fidelity metric's own well-known exponential contraction toward zero for two generically different unitaries as n grows – distinguishing the two is exactly the kind of question the separate, forthcoming empirical results paper is responsible for answering, not this survey.

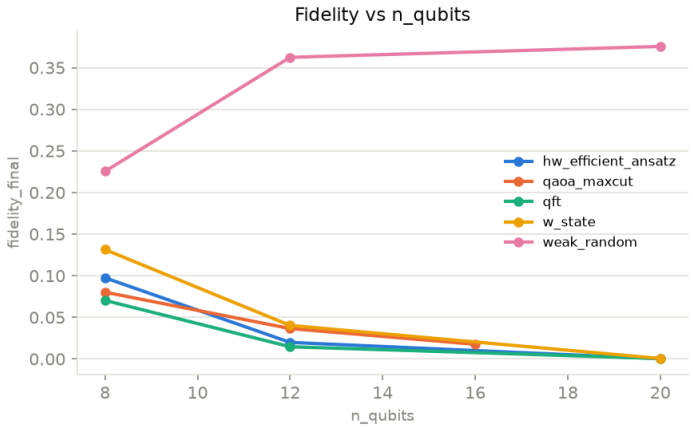


Figure 4: Mean `fidelity_final` vs. qubit count, per circuit family, pooled across the three per-block optimizers of the 225-run core baseline (Section 7.2). Four of five families decline toward the metric's near-zero floor as n grows; `weak_random` is the exception. A single-scale evaluation of any one family in isolation – the norm in the QIEA-MOO literature surveyed above – would surface only one point on this plot.

More fundamentally, the Pareto-dominance relation every classical MOEA baseline in Section 3.3 and every NSGA-II-hybridized QIEA-MOO method in Section 4.5 relies on has itself only recently been shown to have a structural weakness – Zheng et al. [62]'s "weak Pareto boundary" result – that postdates nearly all of the surveyed QIEA-MOO literature and so could not have been accounted for by it. Scalability claims in this field, in short, rest on baselines and dominance relations whose own limits are still being discovered, on top of application-side evaluations that rarely test more than one problem scale to begin with.

7 Case Study: Block-Partitioned Quantum Circuit Optimization

This section grounds several of the preceding sections' more abstract taxonomy and critical-analysis claims in a concrete worked example: this project's own block-partitioned quantum-circuit-optimization pipeline, itself a direct extension of Ghlib et al. [18]'s scalable, block-partitioned NSGA-II framework, adding NSGA-III and SMS-EMOA as alternative per-block optimizers on top of it. As stated in Section 1, the case study is illustrative, not this survey's primary evidence – full empirical validation of the pipeline, including the ablations Sections 7.3 and 7.4 discuss the status of, is the subject of a separate, forthcoming empirical results paper. Every number cited below has been re-verified on the pipeline's current code; where an ablation has not yet been re-run on that code, this section says so explicitly rather than citing a superseded figure, for reasons Section 6.3 already sets out in general terms.

7.1 Pipeline Overview

The pipeline takes a target quantum circuit (or unitary), partitions its qubits into blocks via Louvain community detection on a qubit-interaction graph, evolves a cheaper per-block replacement circuit with a pluggable multi-objective optimizer, and reassembles the blocks into a global circuit. Concretely, it runs in six stages: (1) *partitioning*, which also extracts the gates that cross block boundaries so they can be reinjected later; (2) *sub-circuit extraction*, pulling each block into a self-contained, locally reindexed circuit; (3) a *fidelity metric* comparing the recomposed circuit against the target – exact below a qubit-count threshold, Monte-Carlo-approximate above it, since exact fidelity computation is intractable much beyond roughly a dozen qubits; (4) *intra-block optimization*, where a multi-objective genetic algorithm evolves each block's replacement circuit to jointly maximize fidelity and minimize depth and gate cost; (5) *inter-block gate injection*, re-adding cross-block entanglement that block-local optimization necessarily loses; and (6) *compression*, a mix of hand-rolled and standard circuit-simplification passes applied to the final result.

The pipeline is exercised on five parametric benchmark circuit families – `weak_random`, `qaoa_maxcut`, `w_state`, `qft`, and `hw_efficient_ansatz` – each evaluated at three qubit counts: n=8, 12, and 20, except `qaoa_maxcut`, whose largest size is n=16. Those upper sizes are per-family ceilings imposed by stage (3), not arbitrary choices: above the exact-fidelity threshold, each family needs its own Monte-Carlo fidelity backend – an echo-test estimator by default, or a per-sample-exact statevector estimator for the families where the echo-test estimator is too slow or too coarse (`qaoa_maxcut`, `qft`) – and, for `w_state` and `hw_efficient_ansatz`, more samples and shots, before the

estimate can be trusted at that size. Two of the six stages recur through the rest of this section: stage (4), where NSGA-II, NSGA-III, and SMS-EMOA are interchangeable per-block optimizers (Section 7.2), and stage (5), where a real correctness bug was found and fixed partway through this project's own empirical work. That bug – gate injection silently considering only one pair of blocks out of every partition, and a barrier-counting error that inflated how many gates a partition appeared to cross – had a confirmed, circuit- and seed-dependent impact on final fidelity once fixed, large enough that every quantitative pipeline result computed before the fix is treated as superseded. It is mentioned here, rather than left as an implementation footnote, because it is itself a concrete instance of the reproducibility-gap problem Section 6.3 raises about the wider QIEA-MOO literature: this bug persisted, undetected, through several weeks of active, tested development on a small, single-team codebase, which is a considerably more favorable setting for catching it than the independently authored, rarely-shared-code literature that section discusses.

Figure 5 shows a small instance of one of the pipeline's five benchmark circuit families, `weak_random` – deliberately sparse in its two-qubit gates and connectivity (most of its structure is single-qubit rotations, with only a handful of two-qubit gates connecting a few qubit pairs, and barriers marking generation layers) – to make stage (1) concrete: it is exactly this kind of loosely-coupled qubit structure that a qubit-interaction graph and Louvain community detection can split into blocks with comparatively little entanglement crossing the cuts, which is what stage (5) then has to re-add.

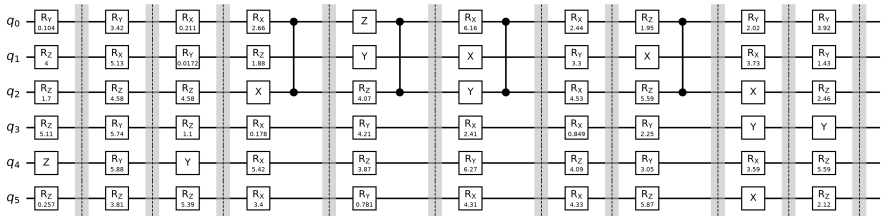


Figure 5: A 6-qubit instance of the `weak_random` benchmark circuit family (Section 7.1), drawn directly from the pipeline's own circuit generator at a small qubit count for legibility. Grey dashed lines mark generation-layer barriers, not two-qubit gates; the only two-qubit gates present are four controlled-Z couplings. This instance's very low connectivity budget allows three candidate qubit pairs (q_0–q_2, q_2–q_3, q_3–q_4), and this particular random seed's independent per-gate draw happened to pick the same pair, q_0–q_2, all four times.

Figure 6 makes stage (1) concrete on a real run rather than only the static circuit above: an 8-qubit `weak_random` instance (seed 0), partitioned by the pipeline's current code into four blocks – $\{0, 2, 4\}$, $\{1\}$, $\{3, 5\}$, and

$\{6, 7\}$. Nodes $\{0, 2, 4, 6, 7\}$ form a single *connected* component in the interaction graph – qubit 4 bridges both halves – yet Louvain splits it into two separate blocks, $\{0, 2, 4\}$ and $\{6, 7\}$, rather than only ever cutting at already-disconnected points; qubit 1, with no interaction edges at all, becomes its own singleton block. The edge crossed by that $\{0, 2, 4\}/\{6, 7\}$ split is exactly the kind of inter-block gate stage (5)'s injection has to reconsider, and stage (1)'s own partitioning choice – not just the circuit's raw connectivity – determines how much of that reconsideration is needed.

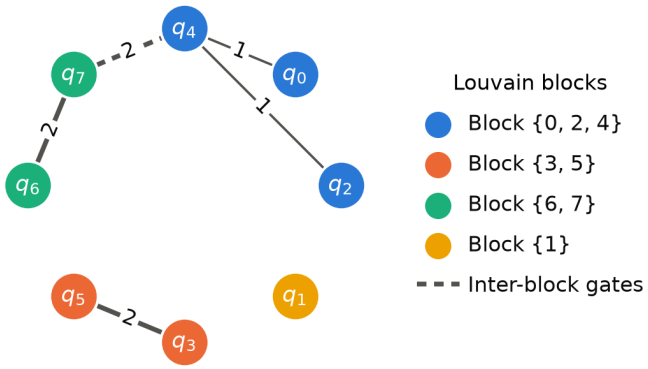


Figure 6: Qubit-interaction graph for the same `weak_random` family (8 qubits, seed 0, generated by running the pipeline's current code, not the static example above) – edge labels are 2-qubit gate counts, the weights Louvain community detection partitions on. Node colour gives each qubit's Louvain block; the dashed edge is the only one crossing a block boundary. Qubits are placed on a circle grouped by block, so position carries no distance meaning.

7.2 NSGA-II vs. SMS-EMOA vs. NSGA-III as a Worked Example

The pipeline's per-block optimizer is selected through a small registry of interchangeable functions sharing one call signature, which makes it a concrete demonstration that the taxonomy's selection axis (Section 4.4) is a genuine, swappable design dimension rather than only an abstract classification: the same partitioning, fidelity metric, injection, and compression stages sit around whichever of NSGA-II's crowding-distance selection, SMS-EMOA's hypervolume-driven environmental selection, or NSGA-III's reference-point niching is plugged in underneath. This is also, incidentally, the reverse pairing already flagged in Sections 4.5 and 6.1: it demonstrates the same three selection paradigms interchanged around one representation, but a domain-

specific, non-quantum-inspired one, not the Q-bit representation the surveyed QIEA-MOO literature has paired with several selection paradigms but never with reference-point or hypervolume-driven selection.

Figure 7 shows what one such run actually looks like under the hood, rather than only its aggregate outcome: the hypervolume and spread of block $\{0, 2, 4\}$'s population, tracked every generation, for the same `weak_random` n=8 instance as Figure 6, optimized here with NSGA-II. Both indicators move together for long stretches but neither improves monotonically – hypervolume repeatedly drops sharply and recovers over the run's first two thirds, consistent with crowding-distance selection trading convergence against diversity within the population rather than strictly improving both every generation, before both indicators plateau in the final third of the run. This is the kind of generation-by-generation behavior the aggregate fidelity/hypervolume comparisons below necessarily average away.

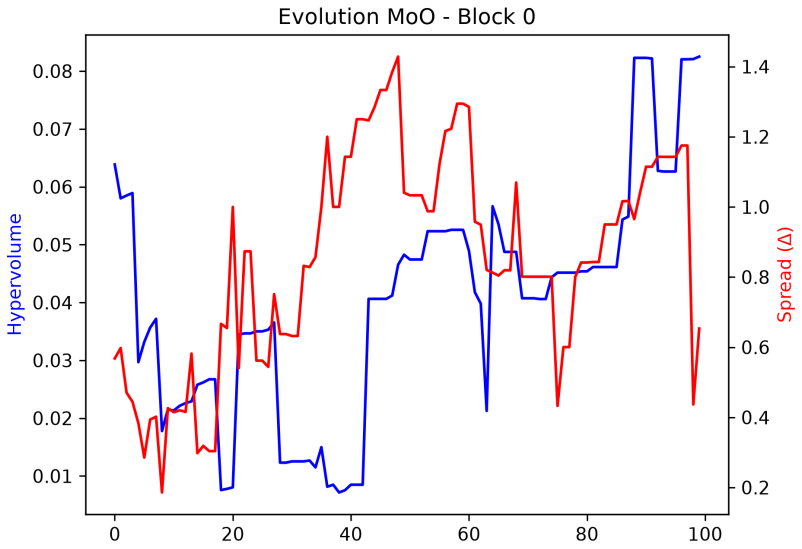


Figure 7: Hypervolume and spread vs. generation for block $\{0, 2, 4\}$ of the same run as Figure 6 (`weak_random`, n=8, seed 0), optimized with NSGA-II over 100 generations, population size 100 (the core baseline's own settings).

Table 6 summarizes the 225-run core baseline re-verified on the fixed injection-stage code (five circuit families, each at the three qubit counts given in Section 7.1, three per-block optimizers, and five seeds per configuration; point mutation, no hybrid local search). NSGA-II reaches the highest mean final fidelity in 13 of the 15 circuit-by-size cells; NSGA-III wins one (`weak_random` at n=8) and SMS-EMOA wins one (`qft` at n=12); Figure 8 shows the pooled means. This mirrors the general shape of an earlier, pre-fix comparison, but the two are not otherwise comparable, since the injection-stage fix changes how much fidelity the pipeline's later stages

recover regardless of which per-block optimizer produced the starting point. On efficiency, upstream of the injection stage and therefore unaffected by that fix, NSGA-III's per-block optimization is cheaper in wall-clock time than NSGA-II's in 14 of the 15 cells – by roughly 1% to 12%, aggregating to about 4.5% cheaper overall – with a single exception (`weak_random` at its own n=20 ceiling, where NSGA-III was about 2% *more* expensive). The direction is consistent with the efficiency gap being caused by the underlying evolutionary library's reference-point selection implementation defaulting to a cheaper non-dominated-sorting routine than the crowding-distance and hand-rolled hypervolume-driven selections use – an implementation detail of this pipeline's current software stack, not a general claim that NSGA-III is more efficient than NSGA-II as algorithms. Nor does it carry over to end-to-end cost: mean total wall-clock per run is roughly 1.9 times *higher* for NSGA-III than for NSGA-II (331s vs. 176s; SMS-EMOA 124s). The whole difference comes from two families at n=20 (`hw_efficient_ansatz` and `w_state`), and none of it from block optimization: about two fifths falls in the injection stage and the rest outside the pipeline's four timed stages. We have not established why. This distinction matters because an earlier measurement in this same project once suggested a much larger, roughly two-fold advantage for NSGA-III; closer, fresher re-measurement showed that figure to be a stale-versus-fresh timing artifact rather than a real effect, and it was retracted once the single-digit-percent figure above was confirmed on repeated measurement. That episode is a direct, self-inflicted instance of the single-run benchmarking weakness Section 6.3 raises about the wider literature – this project is not exempt from the risk it identifies in others, only better positioned to catch it once measured twice.

Table 6: Per-block-optimizer comparison on the 225-run core baseline (5 circuit families × 3 qubit counts × 3 optimizers × 5 seeds; point mutation, no hybrid local search, stochastic injection). "Cells won" counts the circuit-by-size cells (of 15) where that optimizer reached the highest mean `fidelity_final`. Fair hypervolume is defined in Section 7.4.

Per-block optimizer	Mean fidelity_final	Cells won (/15)	Mean block-opt. wall-clock (s)	Mean fair HV
NSGA-II	0.110	13	42.78	0.808
NSGA-III	0.092	1	40.85	0.729
SMS-EMOA	0.092	1	43.88	0.788

7.3 No Universally-Best Mutation Operator

The pipeline exposes a second, independent design axis within each per-block optimizer: the mutation operator applied to a block's chromosome, with

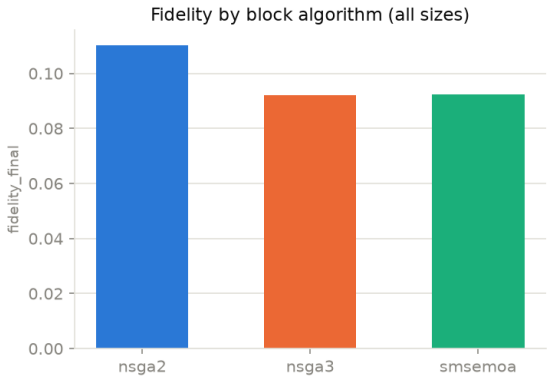


Figure 8: Mean `fidelity_final` by per-block optimizer, pooled across all 15 circuit-by-size cells of Table 6.

point mutation (fixed-length) and two insertion/deletion variants (variable-length) available across all three optimizers in Section 7.2, together with an optional local-search refinement pass applied to the winning individual afterward. An ablation across these mutation schemes and the local-search option, at several problem sizes, was run and analyzed before the injection-stage bug described in Section 7.1 was found, and its headline finding – that no single mutation operator dominates uniformly across problem scale – has not yet been re-run on the fixed code. Because the injection-stage bug changes how much of the final circuit's fidelity this pipeline's later stages recover, and the local-search option's value is entangled with exactly that recovery, that pre-fix finding cannot be assumed to still hold and is deliberately not restated here as a current result, pending that re-run. We report this not as a gap in this section so much as a demonstration of the discipline Section 6.3 argues the wider QIEA-MOO literature should apply more consistently: a finding computed on code later found to contain a real correctness bug is treated as provisional, not citable, until it is re-derived on the corrected code, even when – as here – re-running the ablation is a known, scoped, but not yet completed piece of future work.

7.4 Hypervolume Comparability Across Algorithms

A separate, purely methodological finding from this pipeline's own quality-indicator tooling illustrates the benchmarking-practice limitations raised in Section 6.3 directly, and – unlike Section 7.3's ablation – is structural rather than data-dependent, so it does not need to wait on any re-run to be cited. The pipeline's per-run hypervolume computation derives its reference point and normalization adaptively from that run's own final population, which is appropriate for tracking a single run's own convergence but not valid for

ranking different algorithms or runs against each other, since the same absolute hypervolume value means something different depending on the population it was normalized against. The fix is to also record each run's Pareto front in raw, unnormalized units and, when comparing algorithms, pool all of their front points per block into one shared minimum/maximum normalization and one shared, fixed reference point before computing hypervolume.

Table 7: Adaptive (per-run-normalized) vs. fair (shared-reference) mean hypervolume, by per-block optimizer, at three scale slices of the 225-run core baseline. Fair HV ranks the three optimizers consistently across all three slices; adaptive HV does not rank them consistently even against itself.

	Adaptive mean HV			Fair mean HV		
Scale slice	NSGA-II	NSGA-III	SMS-EMOA	NSGA-II	NSGA-III	SMS-EMOA
All 5 families, all sizes pooled	0.048	0.045	0.055	0.808	0.729	0.788
All 5 families, n=12 (matched)	0.050	0.045	0.054	0.812	0.736	0.808
`qaoa_maxcut` only, its own n=16 ceiling	0.024	0.030	0.028	0.810	0.593	0.739

Table 7 shows both computations, side by side, at three scale slices of the re-verified 225-run core baseline, and Figure 9 plots its pooled row. Adaptive mean HV ranks SMS-EMOA above NSGA-II above NSGA-III when pooling all sizes, but that ranking is not even stable against itself as the scale slice changes: at `qaoa_maxcut`'s own n=16 ceiling, adaptive HV instead ranks NSGA-III above SMS-EMOA above NSGA-II – not a strict reversal of the pooled order (SMS-EMOA and NSGA-II swap places relative to a true reversal), but NSGA-III still moves from worst to best. Fair HV, by contrast, ranks NSGA-II above SMS-EMOA above NSGA-III consistently across all three slices, and NSGA-III's fair-HV gap to NSGA-II widens rather than narrows as problem size grows: 0.052 at n=8 and 0.076 at n=12 (both matched across all five families), 0.084 at n=20 (the four families that reach it), and widest, 0.217, at `qaoa_maxcut`'s own n=16 ceiling. This is a direct, current demonstration of the risk the adaptive-versus-fair distinction warns about: an algorithm-comparison conclusion drawn from adaptive HV alone would not just be questionable in the abstract, it would flip depending on which slice of this pipeline's own data it was drawn from. The methodological point itself, unaffected by any of these specific numbers, is what is load-bearing: cross-algorithm hypervolume comparison requires a shared reference point, and an adaptive, per-run one is only valid for single-run convergence plots.

A second, purely structural guarantee from the same tooling is that the optional local-search refinement pass mentioned in Section 7.3 cannot af-

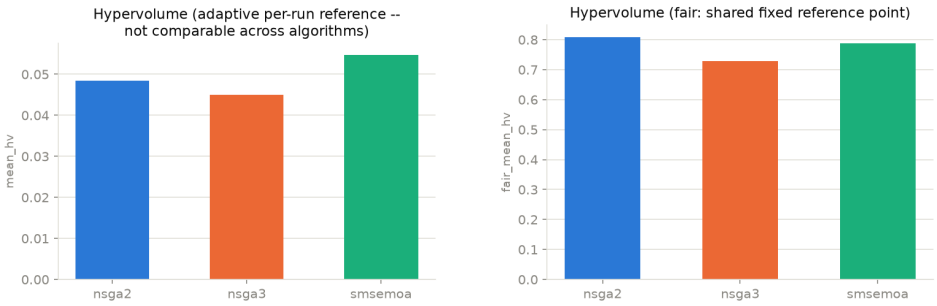


Figure 9: The pooled-all-sizes row of Table 7, shown visually: adaptive mean hypervolume (left, explicitly labeled not comparable across algorithms) ranks SMS-EMOA highest, while fair mean hypervolume (right) ranks NSGA-II highest and NSGA-III lowest – a visibly different ranking from the same underlying Pareto fronts.

fect the recorded Pareto front at all, by construction: the front is captured from the population before that refinement runs, and the refinement only modifies a copy of the winning individual afterward. Fair and adaptive hypervolume are therefore always identical between local-search on and off, at any problem size – a claim about how the indicator is computed, not about any experimental outcome, and one this project has not attempted to unsettle by re-measurement because there is nothing in the fix above that could change it. One question Table 7 raises but does not answer – why NSGA-III's front is not only smaller than NSGA-II's or SMS-EMOA's under the shared-reference comparison but increasingly so as problem size grows, a question about Deb and Jain's reference-point niching versus crowding-distance-based and hypervolume-driven diversity rather than about measurement artifacts – remains open, and we return to it as a candidate direction in Section 8.

8 Open Challenges and Future Directions

Section 6 looked backward: what the surveyed QIEA-for-MOO literature and this survey's own case study already show to be weak or under-tested. This section looks forward, organized into four groups – the two open questions this survey's own analysis raised without answering, methodological directions for QIEA-MOO as a field, technical extensions specific to the case-study pipeline, and the broader trajectory the field appears to be moving toward.

8.1 Open Questions Raised by This Survey

Two specific questions surfaced earlier in this survey without being resolved, and we restate them here as concrete starting points rather than let them dissolve into the more general directions below.

First, Section 4.6 found the application domains covered by the surveyed literature split into two disjoint clusters – quantum-inspired algorithms applied to classical combinatorial or engineering problems, and classical (non-quantum- inspired) algorithms applied to a genuinely quantum problem domain, as in this survey's own case study. Single-objective QIEAs have been applied to quantum circuits [52–55], but we found no *multi-objective* QIEA applied to a quantum-computing problem domain such as circuit optimization or compilation. Concretely, this would mean encoding a gate sequence as a string of Q-bit amplitude pairs, with rotation-gate (and perhaps entanglement- or interference-inspired) operators evolving that encoding directly, and a multi-objective selection mechanism trading fidelity against depth and gate cost – rather than, as this survey's own case study does, using an ordinary classical chromosome of gate tuples inside a classical MOEA. The single-objective evidence is mixed: Krylov and Lukáč [53], comparing a quantum-encoded evolutionary algorithm for circuit design against a classical GPU-accelerated genetic algorithm, report that the quantum encoding brings some disadvantages relative to classical evolutionary computation, while Chou et al. [54] report more compact circuits than a state-of-the-art synthesis method. Whether a multi-objective version would offer any advantage over either cluster alone (e.g. a more natural search-space geometry for gate sequences, given that both objects are already expressed in the language of qubits and rotations) is, to our knowledge, untested.

Second, Section 7.4 found that NSGA-III's Pareto front is not only smaller than NSGA-II's or SMS-EMOA's under a shared-reference (fair) hypervolume comparison in this project's own pipeline, but increasingly so as problem size grows. We could state *that* finding with confidence; we could not state *why* without further investigation reaching beyond survey scope – specifically, whether it traces to Deb and Jain's reference-point niching providing systematically less selection pressure toward a front's extremes than crowding distance or hypervolume-driven environmental selection do, as this pipeline's block sizes grow, or to some interaction specific to this pipeline's block-partitioned setting that would not generalize to non-partitioned NSGA-III use. Distinguishing these is exactly the kind of question the separate, forthcoming empirical results paper is positioned to pursue, since it requires running controlled ablations this survey does not.

8.2 Methodological Directions for QIEA-MOO

Several of Section 6.1's per-category weaknesses point directly at concrete methodological gaps rather than only diagnosing a problem. The missing selection paradigms (Section 4.4) are the clearest: no surveyed method has paired the Q-bit representation with NSGA-III's reference-point niching, SPEA2's density estimator, or SMS-EMOA's hypervolume-driven selection, although MOEA/D-style decomposition has been shown to work with

it [25]. A QIEA-MOO need not commit to a single paradigm, either: Florea et al. [51] run NSGA-II and SPEA2 side by side, letting each produce a share of every generation set by its recent solution quality, and the same scheme could pair a Q-bit representation with SPEA2-style selection without abandoning NSGA-II's. Which of these pairings would actually work well – and whether any of them would shift the operator-versus-selection balance discussed in Section 4.3 – is entirely open. Relatedly, a genuinely interference-based quantum-inspired operator, as opposed to the rotation-gate-plus-one-bolt-on pattern found throughout the surveyed literature, has yet to be demonstrated in a multi-objective QIEA; the only one we found is Narayanan and Moore [20]'s single-objective interference crossover, which predates the Q-bit chromosome and so would first need adapting to it. A third direction is evaluative rather than architectural: establishing a shared benchmark protocol that compares QIEA-MOO methods against each other, not only each against NSGA-II in isolation (Section 6.1), would be a modest undertaking relative to proposing a new algorithm, but it is precisely the missing piece that would let the field's many NSGA-II-beating claims be weighed against one another; Olvera et al. [11]'s multi-run, hypothesis-tested protocol on a standard benchmark suite is a ready starting template. Finally, on the selection side, Guo and Shan [63]'s weak constraint-Pareto dominance mechanism has no QIEA-MOO application yet and no immediate purchase on this survey's own case study either, since the case study's per-block objectives are soft rather than hard constraints today – but a future QIEA-MOO method or pipeline extension that does introduce a hard feasibility cutoff (e.g. a minimum fidelity threshold treated as a constraint rather than a maximized objective) would have a directly applicable mechanism waiting for it.

8.3 Extensions to the Case-Study Pipeline

A separate set of directions is specific to the quantum-circuit- optimization side of the case study rather than to QIEA-MOO as a field, gathered from related work during this survey's own literature search rather than from the surveyed QIEA-MOO methods themselves. On the per-block optimizer side (Section 7.2), the same `BLOCK_OPTIMIZERS`-style registry that already lets NSGA-II, NSGA-III, and SMS-EMOA slot in interchangeably could take further entries: Ulungu et al. [64]'s MOSA is the least costly next addition, since it extends the pipeline's existing simulated-annealing injection machinery rather than introducing a new mechanism from scratch, followed by Zitzler and Künzli [65]'s IBEA (generalizing SMS-EMOA's hypervolume-driven idea to other indicators) or a decomposition-based method in the style of MOEA/D, with SPEA2 – best cited today alongside Alghouass et al. [59]'s stronger, more recent approximation-guarantee result rather than only the original 2001 paper (Section 6.1) – and Coello Coello and Lechuga

[66]'s MOPSO, a different paradigm entirely, as costlier additions requiring a continuous-relaxation or discrete-PSO adaptation of the pipeline's chromosome encoding. On the objective side, a noise- or crosstalk-aware term [67] could become a fourth per-block objective, extending the pipeline's existing (currently placeholder) crosstalk term in its simulated-annealing injection energy, though this would require a noise model the pipeline does not yet have. On scalability specifically, the pipeline's fidelity estimation already required substantial, family-specific engineering to reach the qubit counts used in Section 7.1. Figure 10 shows why this remains an open cost rather than a solved one: total wall-clock time grows sharply with qubit count for four of the five circuit families – most steeply for `hw_efficient_ansatz` (roughly 22s at n=8 to over 1,000s at n=20) and `w_state` (roughly 25s to 900s), more gently for `qft` and `weak_random` – while `qaoa_maxcut` is non-monotonic (26s, 75s, 43s at n=8, 12, and its own n=16 ceiling respectively), plausibly an artifact of the statevector-backend fidelity settings that Section 7.1 notes are applied per family above certain thresholds rather than a genuine efficiency gain at larger n – a wrinkle we did not investigate further here. Classical-shadow or tensor-network fidelity estimation [68] is a concrete candidate for pushing past where the current echo-test and statevector Monte-Carlo backends become memory- or time-bound.

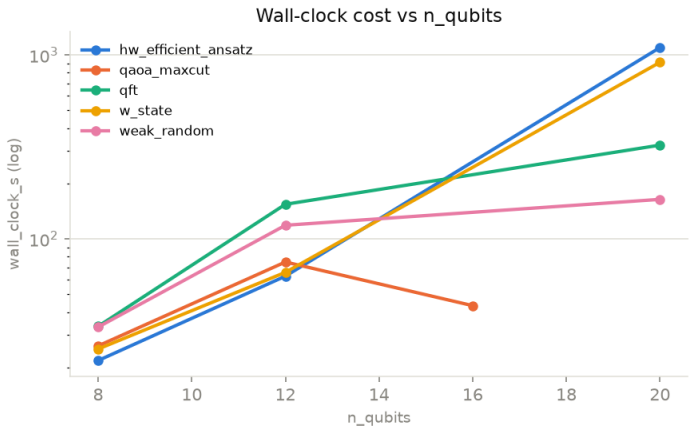


Figure 10: Total wall-clock cost vs. qubit count, per circuit family, pooled across the three per-block optimizers of the 225-run core baseline (log-scaled y-axis). Four families grow sharply with n; `qaoa_maxcut`'s three-point trend is not monotonic (see discussion above).

Finally, two structural extensions target stages this survey's taxonomy does not otherwise cover: reinforcement learning combined with ZX-calculus [69] as an alternative to the pipeline's hand-rolled compression passes, and hyperedge-aware partitioning [70] in place of the fixed-modularity Louvain partitioning Section 7.1 describes. None of these is implemented in the pipeline today;

each is a scoped, independently addable extension rather than a redesign.

8.4 Where the Field Appears to Be Heading

Beyond specific gaps, two recent surveys outside the QIEA-MOO literature proper suggest where multi-objective optimization as a whole is heading, for context rather than as a near-term target for either QIEA-MOO or this survey's case study. Li et al. [71]'s many-objective survey's seven-class taxonomy (relaxed-dominance, diversity-based, aggregation-based, indicator-based, reference-set-based, preference-based, and dimensionality-reduction methods) is a reminder that this survey's own five classical baselines (Section 3.3) sample only some of that landscape. Preference articulation has been tried in a QIEA-MOO only within the MQEA line [24], and we found no QIEA-MOO method built around dimensionality reduction. Zheng and Rasheed [72]'s more recent five-level taxonomy – classical MOEA, ML-guided, RL-driven, LLM-driven, and hybrid neuro-evolutionary methods – places every method surveyed in this paper, and this survey's own case study, at its first level; the field this taxonomy describes is moving toward learned and neural surrogate models and RL- or LLM-guided search rather than purely evolutionary operators, a considerably larger architectural departure than any single direction in Sections 8.2 or 8.3 above. We do not read this as a reason to treat QIEA-MOO or block-partitioned quantum circuit optimization as obsolete – both remain useful, interpretable, comparatively cheap-to-run methods exactly where a learned surrogate's training cost or data requirement is not justified – but a QIEA-MOO method that hybridizes a Q-bit representation with a learned selection or variation mechanism, rather than only with the classical selection paradigms surveyed in Section 4.5, is a direction no method gathered for this survey has yet taken. Finally, Zheng et al. [62]'s structural critique of Pareto-dominance-based EMO (Section 6.3) is itself forward-looking as much as critical: a next generation of QIEA-MOO methods, and of this survey's own case-study pipeline, that addressed the weak Pareto boundary directly – rather than continuing to build on the dominance relation as-is – would be responding to a weakness identified only a year before this survey, not one already absorbed into current practice.

9 Conclusion

Section 1 opened with a gap: quantum-inspired evolutionary algorithms have been applied to multi-objective problems since at least Zhang et al. [9]'s 2003 quantum evolutionary algorithm for multi-objective digital-filter design, yet no prior survey had taxonomized or critically compared QIEA-for-MOO methods on their own terms, separately from single-objective QIEA work on one side and classical MOO on the other. This survey addressed that gap

through the four research questions set out in Section 1.2, preceded by a bibliometric overview.

Field overview: Section 2 mapped 2,250 QIEA papers from 1996–2026. Multi-objective work is under a tenth of the field but its share has roughly doubled since 2024, and in the keyword network it clusters with application domains rather than with the operator and convergence vocabulary of QIEA design – an early signal of the pattern the rest of the survey documents in detail.

RQ1 (classification): Section 4 organized the literature identified in Section 1.1 along five axes – representation scheme, quantum-inspired operator type, selection and diversity mechanism, hybridization with classical MOEAs, and application domain.

RQ2 (comparison): Section 5 applied those axes systematically across the surveyed methods, and in doing so surfaced patterns the taxonomy alone had only implied: selection mechanisms that are varied but never reference-point-based, strength-based, or hypervolume-driven, with the most recent methods converging on NSGA-II's crowding distance; several highly cited methods that do not state how they combine their objectives; and a benchmarking culture in which methods are routinely compared against that same classical baseline but rarely against their own QIEA-MOO siblings.

RQ3 (limitations): Section 6 traced that pattern, and several others like it, into per-category weaknesses – among them a Q-bit representation whose performance degrades as the number of decision variables grows on continuous problems – and three cross-cutting problems – reproducibility gaps, weak benchmarking practices, and scalability claims rarely stress-tested – that recur across the field regardless of which taxonomy category a given method falls into. Comparing the surveyed methods with their single-objective baseline added one more: the rotation gate's reference solution, which has an obvious choice with one objective, is the component that changes most with several, yet only three of the thirteen surveyed methods document how they choose it.

RQ4 (a concrete instance): Section 7 grounded several of these abstract claims in a concrete worked example: this project's own block-partitioned quantum-circuit-optimization pipeline, itself a direct extension of Ghlib et al. [18]'s scalable, block-partitioned NSGA-II framework. We want to restate plainly what that case study was and was not. It was illustrative – a way to show, with a real, actively-maintained codebase, what a swappable selection mechanism looks like in practice, how an adaptive hypervolume computation can silently mislead a cross-algorithm comparison, and how easily a reproducibility gap can hide even inside careful, tested development. It was not this survey's primary evidence, and none of its numbers should be read as a benchmark result for QIEA-MOO methods themselves, which the case study's own pipeline does not use. Full empirical validation of that pipeline – including the mutation-scheme and hybrid-local-search ablations Section 7.3

explicitly left unresolved, pending re-verification on the pipeline's corrected code – is the responsibility of a separate, forthcoming empirical results paper, not this one.

Section 8 closed by turning the critical analysis forward rather than only backward: untested pairings between the Q-bit representation and the reference-point, strength-based, and hypervolume-driven selection paradigms, an interference-based operator not yet demonstrated in the multi-objective setting, concrete extensions to the case-study pipeline drawn from adjacent quantum-computing literature, and the wider field's movement toward learned and neural-guided search that neither QIEA-MOO nor this survey's case study has yet engaged with. None of these close the gap this survey opened with; each narrows it. If the motivation from Section 1 is taken seriously – that a cheaper, shallower optimization process is not merely a performance metric but part of a broader case for energy-efficient computing [16, 17] – then QIEA-MOO's comparative youth as a subfield, documented throughout this survey, is better read as room to grow into than as a reason for caution.

References

[1] Junaid Ur Rehman, Muhammad Shohibul Ulum, Abdurrahman Wachid Shaffar, Amirul Adlil Hakim, Mujirin, Zaid Abdullah, Hayder Al-Hraishawi, Symeon Chatzinotas, and Hyundong Shin. Evolutionary algorithms and quantum computing: Recent advances, opportunities, and challenges. *IEEE Access*, 13:16649–16651, 2025. doi: 10.1109/ACCESS.2025.3530952.

[2] Kuk-Hyun Han and Jong-Hwan Kim. Quantum-inspired evolutionary algorithm for a class of combinatorial optimization. *IEEE Transactions on Evolutionary Computation*, 6(6):580–593, 2002. doi: 10.1109/TEVC.2002.804320.

[3] Yelleti Vivek, Vadlamani Ravi, and P. Radha Krishna. Quantum-inspired evolutionary algorithms for feature subset selection: a comprehensive survey. *Quantum Information Processing*, 24:196, 2025. doi: 10.1007/s11128-025-04787-6.

[4] Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. A fast and elitist multiobjective genetic algorithm: Nsga-ii. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002. doi: 10.1109/4235.996017.

[5] Kalyanmoy Deb and Himanshu Jain. An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part i: Solving problems with box constraints. *IEEE*

Transactions on Evolutionary Computation, 18(4):577–601, 2014. doi: 10.1109/TEVC.2013.2281535.

[6] Eckart Zitzler, Marco Laumanns, and Lothar Thiele. Spea2: Improving the strength pareto evolutionary algorithm. Technical Report 103, Computer Engineering and Networks Laboratory (TIK), ETH Zurich, 2001.

[7] Qingfu Zhang and Hui Li. Moea/d: A multiobjective evolutionary algorithm based on decomposition. *IEEE Transactions on Evolutionary Computation*, 11(6):712–731, 2007. doi: 10.1109/TEVC.2007.892759.

[8] Nicola Beume, Boris Naujoks, and Michael Emmerich. Sms-emoa: Multiobjective selection based on dominated hypervolume. *European Journal of Operational Research*, 181(3):1653–1669, 2007. doi: 10.1016/j.ejor.2006.08.008.

[9] Gexiang Zhang, Weidong Jin, and Laizhao Hu. Quantum evolutionary algorithm for multi-objective optimization problems. In *Proceedings of the 2003 IEEE International Symposium on Intelligent Control (ISIC-03)*, pages 703–708, 2003. doi: 10.1109/ISIC.2003.1254721.

[10] Y. Kim, J.H. Kim, and K.H. Han. Quantum-inspired multiobjective evolutionary algorithm for multiobjective 0/1 knapsack problems. In *Proceedings of the 2006 IEEE Congress on Evolutionary Computation (CEC 2006)*, pages 2601–2606, 2006.

[11] Cynthia Olvera, Oscar Montiel, and Yoshio Rubio. Quantum-inspired evolutionary algorithms on continuous space multiobjective problems. *Soft Computing*, 27:13143–13164, 2023. doi: 10.1007/s00500-022-06916-0.

[12] M. Güzel, F.Y. Okay, İ. Kök, and S. Özdemir. Qnsga-ii: A quantum computing-inspired approach to multi-objective optimization. In *2022 International Symposium on Networks, Computers and Communications (ISNCC)*, 2022. doi: 10.1109/ISNCC55209.2022.9851805.

[13] L. Kumar, S. Solanki, M.K. Jhariya, et al. Quantum-inspired nsga-ii for multi-objective optimization of electric vehicle charging stations. *Scientific Reports*, 16:14666, 2026. doi: 10.1038/s41598-026-44141-z.

[14] Gexiang Zhang. Quantum-inspired evolutionary algorithms: a survey and empirical study. *Journal of Heuristics*, 17:303–351, 2011. doi: 10.1007/s10732-010-9136-0.

[15] Siva Sai, Ishika Goyal, Vinay Chamola, and Rajkumar Buyya. Generative ai in the age of quantum computing: A taxonomy, architectural

elements and future directions. *Future Generation Computer Systems*, 185:108714, 2026. doi: 10.1016/j.future.2026.108714.

[16] Claudia Banciu and Adrian Florea. Aiot at the frontline of climate change management: Enabling resilient, adaptive, and sustainable smart cities. *Climate*, 14(1):19, 2026. doi: 10.3390/cli14010019.

[17] Olimpiu Moga, Claudiu Şolea, Adrian Florea, and Maria Vinţan. Reinforcement learning-based energy management in community microgrids: A comparative study. *Sustainability*, 17(23):10696, 2025. doi: 10.3390/su172310696.

[18] Roumaissa Ghlib, Rania Bouhadouza, and Faicel Hnaien. Scalable multi-objective genetic algorithm for quantum circuit optimization. *Scientific Reports*, 16:17977, 2026. doi: 10.1038/s41598-026-47674-5.

[19] J. David Schaffer. Multiple objective optimization with vector evaluated genetic algorithms. In *Proceedings of the 1st International Conference on Genetic Algorithms*, pages 93–100, 1985.

[20] Ajit Narayanan and Mark Moore. Quantum-inspired genetic algorithms. In *Proceedings of the IEEE International Conference on Evolutionary Computation (ICEC 1996)*, pages 61–66, Nagoya, Japan, 1996.

[21] Kuk-Hyun Han and Jong-Hwan Kim. Quantum-inspired evolutionary algorithms with a new termination criterion, H_ϵ gate, and two-phase scheme. *IEEE Transactions on Evolutionary Computation*, 8(2):156–169, 2004.

[22] Wu Deng, Junjie Xu, Huimin Zhao, and Yingjie Song. A novel gate resource allocation method using improved PSO-based QEA. *IEEE Transactions on Intelligent Transportation Systems*, 23(3):1737–1745, 2022. doi: 10.1109/TITS.2020.3025796.

[23] Bin-Bin Li and Ling Wang. A hybrid quantum-inspired genetic algorithm for multiobjective flow shop scheduling. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, 37(3):576–591, 2007. doi: 10.1109/TSMCB.2006.887946.

[24] Jong-Hwan Kim, Ji-Hyeong Han, Ye-Hoon Kim, Seung-Hwan Choi, and Eun-Soo Kim. Preference-based solution selection algorithm for evolutionary multiobjective optimization. *IEEE Transactions on Evolutionary Computation*, 16(1):20–34, 2012. doi: 10.1109/TEVC.2010.2098412.

[25] Wu Deng, Xing Cai, Daqing Wu, Yingjie Song, Huiling Chen, Xiaojuan Ran, Xiangbing Zhou, and Huimin Zhao. MOQEA/D: Multi-objective QEA with decomposition mechanism and excellent global search and its

application. *IEEE Transactions on Intelligent Transportation Systems*, 25(9):12517–12527, 2024. doi: 10.1109/TITS.2024.3373510.

[26] Mehboob Hussain, Lian-Fu Wei, Fakhar Abbas, Amir Rehman, Muqadar Ali, and Abdullah Raza Lakhan. A multi-objective quantum-inspired genetic algorithm for workflow healthcare application scheduling with hard and soft deadline constraints in hybrid clouds. *Applied Soft Computing*, 128:109440, 2022. doi: 10.1016/j.asoc.2022.109440.

[27] Sajad Mousavi, Fatemeh Afghah, Jonathan Ashdown, and Kurt Turck. Use of a quantum genetic algorithm for coalition formation in large-scale UAV networks. *Ad Hoc Networks*, 87:26–36, 2019. doi: 10.1016/j.adhoc.2018.11.008.

[28] Yong Wang and Dongye Sun. Powertrain matching and optimization of dual-motor hybrid driving system for electric vehicle based on quantum genetic intelligent algorithm. *Discrete Dynamics in Nature and Society*, 2014:956521, 2014. doi: 10.1155/2014/956521.

[29] Jingling Zhang, Wanliang Wang, Yanwei Zhao, and Carlo Cattani. Multiobjective quantum evolutionary algorithm for the vehicle routing problem with customer satisfaction. *Mathematical Problems in Engineering*, 2012:879614, 2012. doi: 10.1155/2012/879614.

[30] Tzyy-Chyang Lu and Gwo-Ruey Yu. An adaptive population multi-objective quantum-inspired evolutionary algorithm for multi-objective 0/1 knapsack problems. *Information Sciences*, 243:39–56, 2013. doi: 10.1016/j.ins.2013.04.018.

[31] Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts, 2022.

[32] Kuk-Hyun Han, Kui-Hong Park, Ci-Ho Lee, and Jong-Hwan Kim. Parallel quantum-inspired genetic algorithm for combinatorial optimization problem. In *Proceedings of the 2001 IEEE Congress on Evolutionary Computation (CEC 2001)*, volume 2, pages 1422–1429, 2001. doi: 10.1109/CEC.2001.934358.

[33] Vincent D. Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10):P10008, 2008. doi: 10.1088/1742-5468/2008/10/P10008.

[34] Zhijin Zhao, Zhen Peng, Shilian Zheng, and Junna Shang. Cognitive radio spectrum allocation using evolutionary algorithms. *IEEE Transactions on Wireless Communications*, 8(9):4421–4425, 2009. doi: 10.1109/TWC.2009.080939.

[35] Jia-Chu Lee, Whei-Min Lin, Gwo-Ching Liao, and Ta-Peng Tsao. Quantum genetic algorithm for dynamic economic dispatch with valve-point effects and including wind power system. *International Journal of Electrical Power & Energy Systems*, 33(2):189–197, 2011. doi: 10.1016/j.ijepes.2010.08.014.

[36] Michael Defoin Platel, Stefan Schliebs, and Nikola Kasabov. Quantum-inspired evolutionary algorithm: A multimodel EDA. *IEEE Transactions on Evolutionary Computation*, 13(6):1218–1232, 2009. doi: 10.1109/TEVC.2008.2003010.

[37] Michael T. M. Emmerich and André H. Deutz. A tutorial on multiobjective optimization: fundamentals and evolutionary methods. *Natural Computing*, 17(3):585–609, 2018. doi: 10.1007/s11047-018-9685-y.

[38] Andrei Pătrăuşanu, Adrian Florea, Mihai Neghină, Alina Dicoiu, and Radu Chiş. A systematic review of multi-objective evolutionary algorithms optimization frameworks. *Processes*, 12(5):869, 2024. doi: 10.3390/pr12050869.

[39] Eckart Zitzler, Lothar Thiele, Marco Laumanns, Carlos M. Fonseca, and Viviane Grunert da Fonseca. Performance assessment of multiobjective optimizers: an analysis and review. *IEEE Transactions on Evolutionary Computation*, 7(2):117–132, 2003. doi: 10.1109/TEVC.2003.810758.

[40] Eckart Zitzler and Lothar Thiele. Multiobjective evolutionary algorithms: a comparative case study and the strength pareto approach. *IEEE Transactions on Evolutionary Computation*, 3(4):257–271, 1999. doi: 10.1109/4235.797969.

[41] Eckart Zitzler. *Evolutionary Algorithms for Multiobjective Optimization: Methods and Applications*. PhD thesis, ETH Zurich, Zurich, Switzerland, 1999.

[42] Thomas Hanne and Mostafa Jabbari Moghaddam. A review of the evolution of multi-objective evolutionary algorithms. *Computers, Materials & Continua*, 85(3):4203–4236, 2025.

[43] Carlos M. Fonseca and Peter J. Fleming. Genetic algorithms for multiobjective optimization: Formulation, discussion and generalization. In *Proceedings of the Fifth International Conference on Genetic Algorithms*, pages 416–423, 1993.

[44] Jeffrey Horn, Nicholas Nafpliotis, and David E. Goldberg. A niched pareto genetic algorithm for multiobjective optimization. In *Proceedings of the First IEEE Conference on Evolutionary Computation, IEEE World Congress on Computational Intelligence*, 1994.

[45] Joshua D. Knowles and David W. Corne. Approximating the nondominated front using the pareto archived evolution strategy. *Evolutionary Computation*, 8(2):149–172, 2000. doi: 10.1162/106365600568167.

[46] David W. Corne, Joshua D. Knowles, and Martin J. Oates. The pareto envelope-based selection algorithm for multiobjective optimization. In *Proceedings of the Sixth International Conference on Parallel Problem Solving from Nature (PPSN VI)*, pages 839–848, 2000.

[47] David W. Corne, Nick R. Jerram, Joshua D. Knowles, and Martin J. Oates. Pesa-ii: Region-based selection in evolutionary multiobjective optimization. In *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO 2001)*, pages 283–290, 2001.

[48] Ye-Hoon Kim and Jong-Hwan Kim. Multiobjective quantum-inspired evolutionary algorithm for fuzzy path planning of mobile robot. In *Proceedings of the 2009 IEEE Congress on Evolutionary Computation (CEC 2009)*, pages 1185–1192, 2009. doi: 10.1109/CEC.2009.4983080.

[49] Si-Jung Ryu, Jong-Hwan Kim, and Ki-Baek Lee. DMQEA: Dual multiobjective quantum-inspired evolutionary algorithm. In *Proceedings of the International Conference on Evolutionary Computation Theory and Applications (ECTA 2014)*, pages 207–214, 2014. doi: 10.5220/0005071802070214.

[50] S. L. Ho, Shiyou Yang, Peihong Ni, and Jin Huang. A quantum-inspired evolutionary algorithm for multi-objective design. *IEEE Transactions on Magnetics*, 49(5):1609–1612, 2013. doi: 10.1109/TMAG.2013.2238661.

[51] Adrian Florea, Ioana Cofaru, Andrei Patrausanu, Nicolae Cofaru, and Ugo Fiore. Superposition of populations in multi-objective evolutionary optimization of car suspensions. *Engineering Applications of Artificial Intelligence*, 126:107026, 2023. doi: 10.1016/j.engappai.2023.107026.

[52] Shengchao Ding, Zhi Hao Jin, and Qing Yang. Evolving quantum circuits at the gate level with a hybrid quantum-inspired evolutionary algorithm. *Soft Computing*, 12(11):1059–1072, 2008. doi: 10.1007/s00500-007-0273-9.

[53] Georgiy Krylov and Martin Lukáč. Quantum encoded quantum evolutionary algorithm for the design of quantum circuits. In *Proceedings of the 16th ACM International Conference on Computing Frontiers (CF '19)*, pages 220–225, 2019. doi: 10.1145/3310273.3322826.

[54] Yao-Hsin Chou, Shu-Yu Kuo, Yu-Chi Jiang, Ching-Hsuan Wu, Jyun-Yi Shen, Cheng-Yen Hua, Pei-Shin Huang, Yun-Ting Lai, Yong Feng

Tong, and Ming-He Chang. A novel quantum-inspired evolutionary computation-based quantum circuit synthesis for various universal gate libraries. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '22)*, pages 2182–2189, 2022. doi: 10.1145/3520304.3533956.

[55] Yaochong Li, Jing Zhang, Ri-Gui Zhou, Yi Qu, and Ruiqing Xu. AQEA-QAS: An adaptive quantum evolutionary algorithm for quantum architecture search. *Entropy*, 27(7):733, 2025. doi: 10.3390/e27070733.

[56] Olivian-Antonio Gîlea, Adrian Florea, Claudia Banciu, and Alexandru Telcean. Scalable quantum inspired evolutionary algorithm for multi-objective optimization real world problems. In *Proceedings of the Euromicro International Conference on Parallel, Distributed, and Network-Based Processing (PDP 2026)*, March 2026. doi: 10.23919/PDP.2026.00054.

[57] Claudia Banciu, Adrian Florea, and M. Popa. Iot-enabled waste collection: Route optimization and environmental impact reduction using smart technologies. In *2025 29th International Conference on System Theory, Control and Computing (ICSTCC)*, pages 782–788, Cluj-Napoca, Romania, 2025. doi: 10.1109/ICSTCC66753.2025.11240462.

[58] Renzhong Deng, Weijie Zheng, and Benjamin Doerr. The first theoretical approximation guarantees for the non-dominated sorting genetic algorithm III (NSGA-III). In *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI 2025)*, pages 8867–8875, 2025. doi: 10.24963/ijcai.2025/986.

[59] Yasser Alghouass, Benjamin Doerr, Martin S. Krejca, and Mohammed Lagmah. Proven approximation guarantees in multi-objective optimization: Spea2 beats nsga-ii. In *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI 2025)*, 2025.

[60] Andrei Pătrăuşanu, Adrian Florea, Mihai Neghină, Radu Chiş, and Alina Dicoiu. A systematic method for generating discrete multidimensional thomae-like functions. *Electronics*, 15(15):3284, 2026. doi: 10.3390/electronics15153284.

[61] Andre Opris, Duc-Cuong Dang, Frank Neumann, and Dirk Sudholt. Runtime analyses of NSGA-III on many-objective problems. In *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO 2024)*, pages 1596–1604, 2024. doi: 10.1145/3638529.3654218.

[62] Ruihao Zheng, Jingda Deng, and Zhenkun Wang. Weak Pareto boundary: A critical challenge for evolutionary multiobjective optimization.

IEEE Transactions on Evolutionary Computation, 30(5):2215–2229, 2026. doi: 10.1109/TEVC.2025.3646963.

[63] Jinhao Guo and Yahui Shan. A constrained multi-objective evolutionary algorithm with weak constraint-pareto dominance and angle distance-based diversity preservation. *Mathematics*, 13(22):3696, 2025. doi: 10.3390/math13223696.

[64] E. L. Ulungu, J. Teghem, Ph. Fortemps, and D. Tuyttens. Mosa method: a tool for solving multiobjective combinatorial optimization problems. *Journal of Multi-Criteria Decision Analysis*, 8(4):221–236, 1999. doi: 10.1002/(SICI)1099-1360(199907)8:4<221::AID-MCDA247>3.0.CO;2-O.

[65] Eckart Zitzler and Simon Künzli. Indicator-based selection in multiobjective search. In *Parallel Problem Solving from Nature (PPSN VIII)*, volume 3242 of *Lecture Notes in Computer Science*, 2004. doi: 10.1007/978-3-540-30217-9_84.

[66] Carlos A. Coello Coello and Maximino Salazar Lechuga. Mopso: A proposal for multiple objective particle swarm optimization. *Proceedings of the 2002 Congress on Evolutionary Computation (CEC'02)*, 2:1051–1056, 2002. doi: 10.1109/CEC.2002.1004388.

[67] Fei Hua, Yuwei Jin, Ang Li, Chenxu Liu, Meng Wang, Yanhao Chen, Chi Zhang, Ari Hayes, Samuel Stein, Minghao Guo, Yipeng Huang, and Eddy Z. Zhang. A synergistic compilation workflow for tackling crosstalk in quantum machines. arXiv preprint arXiv:2207.05751, 2022.

[68] Ge Fu, Hai-Long Zhu, Wen-Bo Chen, Jian Wu, Ding-Fu Zhou, Heng Fan, and Zhang-Qi Yin. Scalable and flexible classical shadow tomography with tensor networks. *Quantum*, 2023.

[69] Jordi Riu, Jan Nogué, Gerard Vilaplana, Artur Garcia-Saez, and Marta P. Estarellas. Reinforcement learning based quantum circuit optimization via zx-calculus. *Quantum*, 2025.

[70] Shane Sweeney and Krishnendu Guha. Dynamic hypergraph partitioning of quantum circuits with hybrid execution. arXiv preprint arXiv:2506.09963, 2025.

[71] Bingdong Li, Jinlong Li, Ke Tang, and Xin Yao. Many-objective evolutionary algorithms: A survey. *ACM Computing Surveys*, 48(1):13, 2015. doi: 10.1145/2792984.

[72] Xufan Zheng and Waqas Rasheed. From pareto to neural: A mathematical survey of multi-objective optimization algorithms—with ap-

plications to software testing. *Mathematics*, 14(15):2694, 2026. doi: 10.3390/math14152694.